A RADIAL–ANGULAR LAPLACE PRINCIPLE FOR HOMOGENEOUS FUNCTIONALS

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Abstract. We study joint upper tails of finitely many positively homogeneous functionals of a random radius and an independent angle. If the angular cost has positive essential infimum and the radius is in the Gumbel domain, regular variation of the angular Laplace integral yields the tail equivalent. The conditional angular law approaches its Gibbs law in total variation and the scaled radial overshoot has an exponential limit independent of any limiting angular statistic. At zero essential infimum, regular variation of the small-cost probabilities and a radial moment condition yield a power-biased limiting radius and an independent Pareto exceedance ratio. We derive explicit equivalents for slack, weakly active and switching constraints.

1. Introduction

Let $R > 0$ be independent of an angle U. For a positively homogeneous functional the event $G_j(RU) > b_j\tau^{\gamma_j}$ requires $R > \tau q_j(U)$ with q_j defined in (2.3). Simultaneous exceedance therefore depends on $q = \max_{1\le j\le d} q_j$. Even for smooth branches q_j this maximum can be nonsmooth at its minimum. For two outputs a slack constraint leaves the usual quadratic approximation unchanged. Under the hypotheses of Theorem 6.1 a tied constraint with zero multiplier restricts the Gaussian integral to a half-space, whereas two positive multipliers yield one linear normal direction. The power in the tail equivalent thus depends on the local behaviour of the branches.

For a smooth scalar functional, Laplace expansions near a minimum manifold yield Weibullian and Gaussian chaos asymptotics [13, Theorems 3 and 5 and Corollary 7]. Gaussian tail and density expansions and a first-order extension to a Gumbel radius with positive continuous angular density are established in [9, Theorems 1.2 and 1.3]. For a positive essential infimum our scalar tail equivalent follows from the random-contraction result of [8]. Theorem 3.1 further identifies the conditional angular law and the exponential overshoot. It applies to angular Laplace equivalents obtained from stratified minima as in [10]. In Proposition 4.1 a logarithmic factor in the tail corresponds to an angular coordinate with a uniform limit on the logarithmic scale.

At zero essential infimum the angular cost can be small enough for the radius to stay on its original scale. In this case Breiman's theorem [2] yields the tail equivalent and Theorem 3.5 identifies the conditional radius and exceedance ratio. The homogeneous densities in [10, Section 10] admit both regimes through an independent generalised-gamma radius.

Section 5 treats local normal forms, including moving thresholds and singular angular densities. We apply the results to a centred Gaussian process (GP) on a finite grid and to squared Brownian norms. Section 8 treats a chi-square example in growing dimension. For smooth Laplace expansions in this setting see [11, 12]. The main proofs are collected in Section 9. The appendices contain criteria for weighted angular laws and the Dirichlet and two-time Brownian calculations.

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2. Exact radial reduction

Let $\mathcal{S}\subset\mathbb{R}^k$ be a compact $n_{\mathcal{S}}$-dimensional C^2 section meeting each ray of the cone exactly once. Let U have probability law μ on $\mathcal{S}$ and let $R>0$ be an independent random variable (rv) with distribution function F, infinite upper endpoint and survival function $\overline{F}(x)=\mathbb{P}\{R>x\}$. Suppose that for $j=1,\dots,d$

$$G_j(ru)=r^{\gamma_j}g_j(u),\qquad r>0,\quad u\in\mathcal{S}, \tag{2.1}$$

where $\gamma_j>0$ and g_j is continuous. Fix $\mathbf{b}=(b_1,\dots,b_d)\in(0,\infty)^d$ and consider the degree-matched thresholds

$$E_\tau(\mathbf{b})=\bigcap_{j=1}^{d}\left\{G_j(RU)>b_j\tau^{\gamma_j}\right\},\qquad \tau\to\infty. \tag{2.2}$$

For $u\in\mathcal{S}$ put

$$q_j(u)=\begin{cases}(b_j/g_j(u))^{1/\gamma_j}, & g_j(u)>0,\\ \infty, & g_j(u)\le 0,\end{cases}\qquad q(u)=\max_{1\le j\le d}q_j(u). \tag{2.3}$$

We use the convention $\overline{F}(\infty)=0$. Assume that the common positivity set $\bigcap_{j=1}^d\{g_j>0\}$ is nonempty. Compactness and lower semicontinuity of the extended costs then imply $0<q_*:=\min_{u\in\mathcal{S}}q(u)<\infty$. Conditioning on U yields the exact identity

$$\mathbb{P}\{E_\tau(\mathbf{b})\}=\int_{\mathcal{S}}\overline{F}\big(\tau q(u)\big)\mu(\mathrm{d}u). \tag{2.4}$$

For $d=2$ we rescale τ so that $\mathbf{b}=(1,b)$ and write $E_\tau(b)=E_\tau((1,b))$.

For an eventually positive threshold vector $\mathbf{t}_\tau=(t_{\tau,1},\dots,t_{\tau,d})$, put

$$\rho_{\tau,j}(u)=\begin{cases}(t_{\tau,j}/g_j(u))^{1/\gamma_j}, & g_j(u)>0,\\ \infty, & g_j(u)\le 0,\end{cases}\qquad \rho_\tau(u)=\max_{1\le j\le d}\rho_{\tau,j}(u), \tag{2.5}$$

and

$$E_\tau(\mathbf{t}_\tau)=\bigcap_{j=1}^{d}\{G_j(RU)>t_{\tau,j}\}. \tag{2.6}$$

Then

$$\mathbb{P}\{E_\tau(\mathbf{t}_\tau)\}=\int_{\mathcal{S}}\overline{F}\big(\rho_\tau(u)\big)\mu(\mathrm{d}u). \tag{2.7}$$

For $t_{\tau,j}=b_j\tau^{\gamma_j}$ we have $\rho_\tau=\tau q$.

Write

$$\mathcal{M}=\{u\in\mathcal{S}:q(u)=q_*\},\qquad r_\tau=\tau q_*. \tag{2.8}$$

We assume that F belongs to the Gumbel max-domain of attraction with auxiliary function w, that is

$$\frac{\overline{F}(r+x/w(r))}{\overline{F}(r)}\longrightarrow e^{-x},\qquad x\in\mathbb{R},\quad r\to\infty, \tag{2.9}$$

see, for example, [4], and set

$$\lambda_\tau=r_\tau w(r_\tau). \tag{2.10}$$

For an infinite-endpoint Gumbel law, $1/w(r)=o(r)$ as $r\to\infty$; hence $\lambda_\tau\to\infty$ as $\tau\to\infty$.

Compactness and lower semicontinuity imply that for every sufficiently small neighbourhood V of $\mathcal{M}$ there is $\eta_V>0$ such that

$$\inf_{u\notin V}\frac{q(u)}{q_*}\ge 1+\eta_V. \tag{2.11}$$

For moving thresholds we shall require a uniform version of (2.11). For every fixed $p > 0$ there is $C_p < \infty$ such that

$$\frac{\overline{F}\big(r_\tau(1 + x/\lambda_\tau)\big)}{\overline{F}(r_\tau)} \le C_p(1 + x)^{-p}, \qquad x \ge 0, \tag{2.12}$$

for all sufficiently large τ. In order to verify this bound and its uniformity in x we use the Gumbel representation [4]

$$\overline{F}(r) = c(r) \exp\left(-\int_{r_0}^{r} \frac{\mathrm{d}t}{a(t)}\right), \qquad c(r) \to c > 0, \quad a(r)w(r) \to 1,$$

as $r \to \infty$, where $a > 0$ is absolutely continuous and $a'(r) \to 0$ as $r \to \infty$. For any $\epsilon > 0$ eventually $a(t) \le a(r) + \epsilon(t - r)$ for every $t \ge r$. Hence

$$\frac{\overline{F}(r + s)}{\overline{F}(r)} \le C\left(1 + \frac{\epsilon s}{a(r)}\right)^{-1/\epsilon}, \qquad s \ge 0.$$

Taking $s = x/w(r)$ and $1/\epsilon > p$ proves (2.12). On $\mathcal{S} \setminus V$ we have $x = \lambda_\tau(q/q_* - 1) \ge \eta_V \lambda_\tau$ by (2.11), so (2.12) bounds the contribution from this set in (2.4). See also the rapid-variation estimates in [3].

3. Two radial–angular transfer regimes

We extend (2.4) to measurable angular costs. A positive essential infimum forces a large radius, whereas a zero infimum permits an exceedance at a bounded radius.

3.1. **A positive essential minimum.** Let $U \sim \mu$ on $(\mathcal{S}, \mathcal{A})$ be independent of R. The radial distribution F has infinite upper endpoint and satisfies (2.9). For a measurable $q : \mathcal{S} \to (0, \infty]$ with $q_* := \operatorname{essinf}_\mu q \in (0, \infty)$ put

$$E_u = \{R > uq(U)\}, \qquad r_u = uq_*, \qquad \lambda_u = r_u w(r_u), \qquad W(v) = \frac{q(v)}{q_*} - 1. \tag{3.1}$$

Take $W \ge 0$ by modification on a μ-null set and define

$$H(s) = \mu\{W \le s\}, \qquad L_{\mathrm{ang}}(z) = \int_{\mathcal{S}} e^{-zW(v)} \mu(\mathrm{d}v), \qquad s \ge 0, \quad z > 0, \tag{3.2}$$

with $e^{-\infty} = 0$. For $K > 0$, $\kappa \ge 0$ and ℓ positive and slowly varying at infinity, Karamata's theorem [1] yields

$$\begin{aligned} L_{\mathrm{ang}}(z) &\underset{z\to\infty}{\sim} K z^{-\kappa} \ell(z) \\ &\iff \quad H(s) \underset{s\downarrow 0}{\sim} \frac{K}{\Gamma(\kappa + 1)} s^\kappa \ell(1/s). \end{aligned} \tag{3.3}$$

For $\kappa = 0$ this includes both $\mu\{W = 0\} > 0$ and $H(0) = 0$. Let Q_z be the probability measure

$$Q_z(\mathrm{d}v) = \frac{e^{-zW(v)}}{L_{\mathrm{ang}}(z)} \mu(\mathrm{d}v). \tag{3.4}$$

On E_u put $B_u = w(r_u)(R - uq(U))$.

Theorem 3.1. *If L_{ang} varies regularly at infinity with index $-\kappa$, $\kappa \ge 0$, then*

$$\mathbb{P}\{E_u\} \underset{u\to\infty}{\sim} \overline{F}(r_u) L_{\mathrm{ang}}(\lambda_u) \underset{u\to\infty}{\sim} \Gamma(\kappa + 1) \overline{F}(r_u) H(1/\lambda_u). \tag{3.5}$$

For each fixed $y \ge 0$

$$\sup_{A \in \mathcal{A}} \left| \mathbb{P}\{U \in A,\ B_u > y \mid E_u\} - e^{-y} Q_{\lambda_u}(A) \right| \xrightarrow{u\to\infty} 0. \tag{3.6}$$

If $T_u : \mathcal{S} \to \mathsf{X}$ is measurable, X is Polish and $Q_{\lambda_u} \circ T_u^{-1} \Rightarrow Q$ as $u \to \infty$, then conditionally on E_u

$$(T_u(U), B_u) \xrightarrow[u\to\infty]{d} (T, E), \qquad T \sim Q, \quad E \sim \mathrm{Exp}(1), \tag{3.7}$$

where T and E are independent.

In the notation of (3.3), (3.5) reads

$$\mathbb{P}\{E_u\} \underset{u\to\infty}{\sim} K\overline{F}(r_u)\lambda_u^{-\kappa}\ell(\lambda_u). \tag{3.8}$$

Taking $y = 0$ in (3.6) yields

$$\sup_{A\in\mathcal{A}} |\mathbb{P}\{U \in A \mid E_u\} - Q_{\lambda_u}(A)| \xrightarrow{u\to\infty} 0. \tag{3.9}$$

For $S = q_*/q(U)$ the random-contraction equivalent [8, Theorem 3.1(a)] yields (3.5) with endpoint atoms handled directly. Equation (3.6) compares finite angular measures at each y, while (3.7) yields joint weak convergence.

Corollary 3.2. *Under Theorem 3.1 let $A_u = \lambda_u W(U)$. Conditionally on E_u*

$$(A_u, B_u) \xrightarrow[u\to\infty]{d} (G_\kappa, E), \tag{3.10}$$

where $G_\kappa \sim \mathrm{Gamma}(\kappa, 1)$ and $E \sim \mathrm{Exp}(1)$ are independent. We use the rate parametrisation and set $G_0 = 0$. Moreover

$$w(r_u)(R - r_u) \;\Big|\; E_u \xrightarrow[u\to\infty]{d} \mathrm{Gamma}(\kappa + 1, 1) \tag{3.11}$$

and with $R/\infty = 0$

$$q_* w(r_u)\left(\frac{R}{q(U)} - u\right) \;\Big|\; E_u \xrightarrow[u\to\infty]{d} \mathrm{Exp}(1). \tag{3.12}$$

For the costs in (2.3) the following identity holds on the common positivity set for arbitrary degrees $\gamma_j > 0$

$$\frac{R}{q(U)} = \min_{1\le j\le d}\left(\frac{G_j(RU)}{b_j}\right)^{1/\gamma_j}.$$

Remark 3.3. *Let $C_1, \dots, C_m$ be disjoint measurable cells with*

$$\int_{C_a} e^{-zW(v)}\mu(\mathrm{d}v) \underset{z\to\infty}{\sim} K_a z^{-\kappa_a}, \qquad K_a > 0, \quad \kappa_a \ge 0,$$

and suppose that the integral over their complement is $o(z^{-\kappa_*})$ as $z \to \infty$, where $\kappa_* := \min_{1\le a\le m} \kappa_a$. Then

$$\mathbb{P}\{U \in C_a \mid E_u\} \xrightarrow{u\to\infty} \begin{cases} K_a \Big/ \displaystyle\sum_{b:\kappa_b=\kappa_*} K_b, & \kappa_a = \kappa_*, \\ 0, & \kappa_a > \kappa_*. \end{cases}$$

For cells with positive limiting weight, (3.9) also transfers coordinate limits conditional on the cell.

Remark 3.4. *For fixed q and μ, (3.8) includes logarithmic factors. Higher-order equivalents require radial error bounds and moving costs or increasing dimension require uniform estimates.*

3.2. **A zero essential minimum.** Let $U \sim \mu$ on a measurable space $\mathcal{U}$ be independent of $R > 0$ and let $q : \mathcal{U} \to (0, \infty]$ be measurable with $\operatorname{essinf}_\mu q = 0$. Set

$$E_u^0 := \{R > uq(U)\}, \qquad H_0(s) := \mathbb{P}\{q(U) < s\}, \quad s > 0. \tag{3.13}$$

The strict sublevel ensures $\mathbb{P}\{E_u^0\} = \mathbb{E}\{H_0(R/u)\}$ also for laws with atoms. Write $m_\alpha = \mathbb{E}\{R^\alpha\}$.

Theorem 3.5. *Suppose $\mathbb{E}\{R^{\alpha+\varepsilon}\} < \infty$ for some $\alpha, \varepsilon > 0$ and*

$$H_0(1/u) \underset{u\to\infty}{\sim} C_0 u^{-\alpha} \ell_0(u) \tag{3.14}$$

with $C_0 > 0$ and ℓ_0 positive and slowly varying at infinity. Then

$$\mathbb{P}\{E_u^0\} \underset{u\to\infty}{\sim} m_\alpha H_0(1/u) \underset{u\to\infty}{\sim} C_0 m_\alpha u^{-\alpha} \ell_0(u). \tag{3.15}$$

For independent R_α and B_α with $\mathbb{P}\{B_\alpha \le t\} = t^\alpha$ on $[0,1]$ and

$$\mathbb{P}\{R_\alpha \in \mathrm{d}r\} = \frac{r^\alpha}{m_\alpha} \mathbb{P}\{R \in \mathrm{d}r\}, \qquad r > 0, \tag{3.16}$$

we have conditionally on E_u^0

$$\left(R, \frac{uq(U)}{R}\right) \xrightarrow[u\to\infty]{d} (R_\alpha, B_\alpha). \tag{3.17}$$

Equivalently, with $P_\alpha = 1/B_\alpha$

$$\left(R, \frac{R}{uq(U)}\right) \,\Big|\, E_u^0 \xrightarrow[u\to\infty]{d} (R_\alpha, P_\alpha), \qquad \mathbb{P}\{P_\alpha > v\} = v^{-\alpha}, \quad v \ge 1. \tag{3.18}$$

Breiman's theorem [2] applied to $R/q(U)$ yields (3.15). The conditional radius has the power-biased law (3.16) in the limit.

Remark 3.6. *An infinite-endpoint Gumbel law satisfies the moment condition in Theorem 3.5 for every $\alpha > 0$ by rapid variation of its tail [4].*

3.3. **Homogeneous densities and the two cost regimes.** Consider the density of [10, Section 10]

$$p_\eta(v) = A_0(v) e^{-V(v)}, \qquad v \in \mathbb{R}^D, \qquad A_0(rv) = r^{\beta_A} A_0(v), \quad V(rv) = r^\beta V(v), \quad r > 0.$$

Here A_0, V are nonnegative and measurable, $\beta_A > -D$ and $\beta > 0$. Write σ for surface measure on $\mathbb{S}^{D-1}$. We shall assume that $V > 0$ σ-almost everywhere on $\{A_0 > 0\}$ and that $W = A_0/V^a$ is integrable, where $a = (D + \beta_A)/\beta$ and $W = 0$ off $\{A_0 V > 0\}$. Put

$$\Theta = \eta/\|\eta\|, \qquad R_0 = \|\eta\| V(\Theta)^{1/\beta}.$$

Proposition 3.7. *R_0 and Θ are independent with*

$$\begin{aligned} R_0^\beta &\sim \mathrm{Gamma}(a, 1), \qquad \eta = R_0 \frac{\Theta}{V(\Theta)^{1/\beta}}, \\ \mu_V(\mathrm{d}\theta) &:= \mathbb{P}\{\Theta \in \mathrm{d}\theta\} = \frac{\Gamma(a)}{\beta} W(\theta) \mathrm{d}\sigma(\theta). \end{aligned} \tag{3.19}$$

For $G_j(r\theta) = r^{\gamma_j} g_j(\theta)$ with $b_j, \gamma_j > 0$ put

$$q_V(\theta) = V(\theta)^{1/\beta} \max_{1\le j\le d} \left(\frac{b_j}{g_j(\theta)}\right)^{1/\gamma_j}, \qquad \Phi_V = q_V^\beta, \tag{3.20}$$

taking $q_V = \infty$ if any $g_j(\theta) \le 0$. Then for $\tau > 0$

$$\mathbb{P}\left\{\bigcap_{j=1}^d \{G_j(\eta) > b_j \tau^{\gamma_j}\}\right\} = \mathbb{P}\{R_0 > \tau q_V(\Theta)\}$$

$$= \frac{1}{\beta}\int_{\mathbb{S}^{D-1}} W(\theta)\Gamma\big(a, \tau^\beta \Phi_V(\theta)\big)\mathrm{d}\sigma(\theta), \tag{3.21}$$

where $\Gamma(a,y)=\int_y^\infty t^{a-1}e^{-t}\mathrm{d}t$ *and* $\Gamma(a,\infty)=0$.

Proof. In polar coordinates $t = r^\beta V(\theta)$ yields the joint density $\beta^{-1}W(\theta)t^{a-1}e^{-t}$ of (Θ, R_0^β), whence (3.19). By homogeneity the joint event is $\{R_0 > \tau q_V(\Theta)\}$. Integrating the gamma tail yields (3.21). □

For a positive essential minimum of q_V, Theorem 3.1 applies under its angular hypotheses with $w_0(r) = \beta r^{\beta-1}$. A zero minimum can arise when the section $V = 1$ or its outputs are unbounded.

Corollary 3.8. *In the setting of Proposition 3.7 assume that for some* $c_0, h > 0$

$$\mu_V\{\Phi_V \le s\} \underset{s\downarrow 0}{\sim} c_0 s^h. \tag{3.22}$$

Then

$$\mathbb{P}\{R_0 > \tau q_V(\Theta)\} \underset{\tau\to\infty}{\sim} c_0 \frac{\Gamma(a+h)}{\Gamma(a)}\tau^{-\beta h} \tag{3.23}$$

and conditionally on this event

$$\left(R_0^\beta, \frac{\tau q_V(\Theta)}{R_0}\right) \xrightarrow[\tau\to\infty]{d} (T_h, B_{\beta h}), \tag{3.24}$$

where $T_h \sim \mathrm{Gamma}(a+h,1)$ *and* $B_{\beta h} \sim \mathrm{Beta}(\beta h, 1)$ *are independent.*

Proof. By (3.22), comparison with the closed sublevels at $(1-\delta)s$ and s yields $\mu_V\{q_V < s\} \sim c_0 s^{\beta h}$ as $s \downarrow 0$ on letting $\delta \downarrow 0$. Since $R_0^\beta \sim \mathrm{Gamma}(a,1)$, all radial moments exist and $\mathbb{E}\left\{R_0^{\beta h}\right\} = \Gamma(a+h)/\Gamma(a)$. Apply Theorem 3.5 with $\alpha = \beta h$. Weighting the gamma density by t^h changes its shape to $a+h$. □

Under (3.22), (3.24) implies conditional tightness of $\tau^\beta \Phi_V(\Theta)$. If

$$\int_{\{\Phi_V \le s\}} W(\theta)\mathrm{d}\sigma(\theta) \underset{s\downarrow 0}{\sim} K s^h$$

for $K > 0$ then (3.19) yields $c_0 = \Gamma(a)K/\beta$ and the coefficient in (3.23) is $K\Gamma(a+h)/\beta$. Under the interior-zero hypotheses of [10, Section 10] the normal model G_p has degree one under dilations of volume degree Q, whence

$$h = Q, \qquad K = \int_{\mathcal{Z}} W(p)\,\mathrm{vol}\{t : G_p(t) \le 1\}\mathrm{d}V_{\mathcal{Z}}(p).$$

Under the fractional-frontier hypotheses there, $\Phi_V(t,v) \sim c(t)v^{1-\zeta}$ as the inward coordinate $v \downarrow 0$, with $0 < \zeta < 1$. Writing ω for W times the collar Jacobian yields

$$h = \frac{1}{1-\zeta}, \qquad K = \int_{\mathcal{B}} \omega(t,0)c(t)^{-h}\mathrm{d}V_{\mathcal{B}}(t).$$

Both formulas require the respective geometry, angular gap and integrability assumptions. If the limiting angular weight vanishes identically then $K = 0$ and the leading equivalent needs further analysis.

Example 3.9. *Let* X *have density* $(p+1)x^p$ *on* $(0,1)$ *with* $p > -1$ *and let* $R > 0$ *be independent of* X. *For* $k > 0$ *put* $\alpha = (p+1)/k$. *Since* $\mathbb{P}\{X^k \le s\} = s^\alpha$ *for* $0 \le s \le 1$

$$\mathbb{P}\left\{R > \tau X^k\right\} = \mathbb{E}\{\min((R/\tau)^\alpha, 1)\}. \tag{3.25}$$

If $m_\alpha = \mathbb{E}\{R^\alpha\} < \infty$ *then*

$$\mathbb{P}\Big\{R > \tau X^k\Big\} \underset{\tau\to\infty}{\sim} m_\alpha \tau^{-\alpha}, \qquad \left(R, \frac{\tau X^k}{R}\right) \,\Big|\, \{R > \tau X^k\} \xrightarrow[\tau\to\infty]{d} (R_\alpha, B_\alpha)$$

with the independent limits of Theorem 3.5. Indeed for $0 < v \le 1$ *and bounded continuous* f *the same conditioning yields*

$$\tau^\alpha \mathbb{E}\Big\{f(R)\mathbf{1}_{\{\tau X^k < vR\}}\Big\} = \mathbb{E}\{f(R)\min((vR)^\alpha, \tau^\alpha)\} \xrightarrow[\tau\to\infty]{} v^\alpha \mathbb{E}\{f(R)R^\alpha\}.$$

For $R^\beta \sim \mathrm{Gamma}(a, 1)$ *we have* $m_\alpha = \Gamma(a+\alpha/\beta)/\Gamma(a)$ *and* $R_\alpha^\beta \sim \mathrm{Gamma}(a+\alpha/\beta, 1)$. *On an angular arc* $\theta(x)$ *the degree-*γ *output* $G(r\theta(x)) = r^\gamma x^{-k\gamma}$, $\gamma > 0$, *realises the cost* $q(\theta(x)) = x^k$ *at unit threshold.*

4. Stratified angular localisation

For $q_* > 0$ the cellwise estimates in [10] apply through Theorem 3.1 once they establish regular variation of L_{ang} and convergence of the rescaled Gibbs laws. The decomposition must account for overlaps and for the mass outside the cells. For finitely many power contributions the limiting weights are those of Remark 3.3.

4.1. A logarithmic collar for two simultaneous outputs.

We realise the critical collar of [10] with $m = 2$ and $k = 3/2$ as the maximum of two smooth costs. For $0 < \varepsilon < 1$ put

$$\Psi(x, y) = \frac{(x, y, 1)}{\sqrt{1 + x^2 + y^2}}, \qquad (x, y) \in D := (0, \varepsilon) \times (-\varepsilon, \varepsilon). \tag{4.1}$$

Let $U = \Psi(X, Y)$ where (X, Y) has bounded density h on D with

$$\lim_{D\ni(x,y)\to(0,0)} h(x, y) = h_0 > 0. \tag{4.2}$$

For $q_* > 0$ define

$$\begin{aligned} q_\pm(\Psi(x, y)) &= q_*\big(1 + x^2y^2 \pm y^3\big), \\ q = \max(q_+, q_-) &= q_*\big(1 + \psi(x, y)\big), \qquad \psi(x, y) = x^2y^2 + |y|^3. \end{aligned} \tag{4.3}$$

Since $\varepsilon < 1$, both costs are positive on the closed patch. A common smooth cutoff to $2q_*$ extends them positively to $\mathbb{S}^2$ with $q \ge q_*$. For $\gamma > 0$ set

$$G_\pm(r\theta) = \left(\frac{r}{q_\pm(\theta)}\right)^\gamma, \qquad r > 0, \quad \theta \in \mathbb{S}^2. \tag{4.4}$$

Let R be independent of U with distribution satisfying (2.9) and put

$$\begin{aligned} E_u &= \{G_+(RU) > u^\gamma,\ G_-(RU) > u^\gamma\} = \{R > uq(U)\}, \\ r_u &= uq_*, \qquad \lambda_u = r_u w(r_u). \end{aligned} \tag{4.5}$$

Write $B_u \in \{+, -\}$ for the index of the smaller output, breaking ties arbitrarily.

Proposition 4.1. *With* $c = h_0\sqrt{\pi}/6$ *we have*

$$L(z) := \int_D e^{-z\psi(x,y)} h(x, y)\mathrm{d}x\mathrm{d}y \underset{z\to\infty}{\sim} cz^{-1/2}\log z, \tag{4.6}$$

$$\mathbb{P}\{E_u\} \underset{u\to\infty}{\sim} c\overline{F}(r_u)\lambda_u^{-1/2}\log\lambda_u. \tag{4.7}$$

For

$$T_u = -\frac{\log X}{\log\lambda_u}, \qquad V_u = \sqrt{\lambda_u}XY, \qquad Z_u = w(r_u)(R - uq(U)) \tag{4.8}$$

the following limits hold jointly under conditioning on E_u

$$(T_u, V_u, Z_u) \xrightarrow[u\to\infty]{d} (T, V, Z), \tag{4.9}$$

$$B_u \xrightarrow[u\to\infty]{d} B = \begin{cases} +, & V>0, \\ -, & V<0. \end{cases} \tag{4.10}$$

Here $T \sim \mathrm{Unif}(0,1/6)$, $V \sim N(0,1/2)$ *and* $Z \sim \mathrm{Exp}(1)$ *are independent. Hence* B *is a fair sign independent of* (T,Z).

For fixed $x>0$ the quadratic term x^2y^2 yields a Gaussian integral proportional to $x^{-1}z^{-1/2}$. Integration towards $x=0$ produces the logarithm in (4.6). The uniform limit in (4.9) describes the resulting range of angular scales.

5. Normal forms for finitely many outputs

Let $q_* = \min_{u\in\mathcal{S}} q(u)$ and let $\mathcal{M}$ be the minimising set in (2.8). We shall assume that $\mathcal{M}$ is a compact boundaryless C^2 submanifold of $\mathcal{S}$ of dimension m and codimension $c = n_{\mathcal{S}} - m \geq 1$. Write

$$\mathsf{N}\mathcal{M} = \{(s,x) : s\in\mathcal{M},\ x\in N_s\mathcal{M}\}, \qquad f_j = q_j/q_*, \qquad f = \max_{1\leq j\leq d} f_j = q/q_*.$$

Here $N_s\mathcal{M}$ is the normal space in $\mathcal{S}$ and the bundle volume is $\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x$. Fix a tubular chart $\Psi(s,z)$ on $\mathcal{T}$ with volume Jacobian one at $z=0$. Suppose μ has a positive continuous density h near $\mathcal{M}$ with respect to Riemannian volume.

For $\lambda\geq 1$ let $D_{\lambda,s} : N_s\mathcal{M} \to N_s\mathcal{M}$ be a continuous family of invertible linear maps satisfying

$$|\det D_{\lambda,s}| = \lambda^{-\kappa}, \qquad \kappa>0, \tag{5.1}$$

and

$$\sup_{s\in\mathcal{M}} \|D_{\lambda,s}\| \longrightarrow 0, \qquad \lambda\to\infty. \tag{5.2}$$

On the chart domain suppose that

$$\lambda\left[f\big(\Psi(s,D_{\lambda,s}x)\big) - 1\right] \xrightarrow{\lambda\to\infty} \Phi_s(x), \tag{5.3}$$

for bundle-volume almost every (s,x), where $(s,x)\mapsto\Phi_s(x)\in[0,\infty]$ is Borel. By (5.2) the domain eventually contains each compact subset of $\mathsf{N}\mathcal{M}$. For some $c_0>0$ and $p>\kappa$ suppose there are jointly Borel, uniformly coercive gauges $\varrho_s : N_s\mathcal{M}\to[0,\infty)$ such that

$$\lambda\left[f\big(\Psi(s,D_{\lambda,s}x)\big) - 1\right] \geq c_0\varrho_s(x), \tag{5.4}$$

$$\sup_{s\in\mathcal{M}} \int_{N_s\mathcal{M}} (1+\varrho_s(x))^{-p}\mathrm{d}x < \infty, \tag{5.5}$$

where (5.4) holds throughout the chart domain. With $e^{-\infty}=0$ the following integral is finite. We shall assume

$$C_\Phi := \int_{\mathcal{M}} h(s) \int_{N_s\mathcal{M}} e^{-\Phi_s(x)}\mathrm{d}x\mathrm{d}V_{\mathcal{M}}(s) > 0. \tag{5.6}$$

On $\{U\in\mathcal{T}\}$ write $U=\Psi(S_\tau,z_\tau)$ and extend these coordinates arbitrarily off the tube.

Theorem 5.1. *Under the preceding assumptions and* (2.9) *we have*

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau\to\infty}{\sim} \overline{F}(r_\tau)\lambda_\tau^{-\kappa}C_\Phi. \tag{5.7}$$

Conditionally on $E_\tau(\mathbf{b})$, U *lies in* $\mathcal{T}$ *with probability tending to one and the rescaled coordinates converge in total variation*

$$\left(S_\tau, D^{-1}_{\lambda_\tau,S_\tau} z_\tau\right) \xrightarrow[\tau\to\infty]{d} \mathsf{Q}_\Phi, \qquad \mathsf{Q}_\Phi(\mathrm{d}s,\mathrm{d}x) = C_\Phi^{-1}h(s)e^{-\Phi_s(x)}\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x. \tag{5.8}$$

Jointly with (5.8) *the radial excess satisfies*

$$w(r_\tau)\big(R-\tau q(U)\big) \ \Big|\ E_\tau(\mathbf{b}) \xrightarrow[\tau\to\infty]{d} \mathrm{Exp}(1), \tag{5.9}$$

with a limit independent of the angular coordinates.

For $d \geq 2$, $\gamma_1 = \cdots = \gamma_d = \gamma$ and $\mathbf{b} = \mathbf{1}_d$ put

$$M_\tau = \min_{1\leq j\leq d} G_j(RU), \qquad J_\tau = \min \operatorname*{argmin}_{1\leq j\leq d} G_j(RU).$$

Let $(S, X) \sim \mathsf{Q}_\Phi$ and $E \sim \mathrm{Exp}(1)$ be independent.

Corollary 5.2. *Under Theorem 5.1 and the common degree and threshold above, put*

$$\zeta_\tau := \frac{q_* w(r_\tau)}{\gamma \tau^{\gamma-1}} = \frac{\lambda_\tau}{\gamma \tau^\gamma}. \tag{5.10}$$

Conditionally on $E_\tau(\mathbf{1}_d)$ we have

$$\zeta_\tau (M_\tau - \tau^\gamma) \xrightarrow[\tau\to\infty]{d} E. \tag{5.11}$$

For $\lambda \geq 1$ define on the tubular domain

$$\boldsymbol{\varphi}_\lambda(s, x) := \big(\lambda \left[f_j\big(\Psi(s, D_{\lambda,s}x)\big) - 1\right]\big)_{j=1}^d. \tag{5.12}$$

If for a Borel map $\boldsymbol{\varphi} = (\varphi_{j,s})_{j=1}^d$ into $[-\infty, \infty]^d$

$$\boldsymbol{\varphi}_\lambda(s, x) \xrightarrow{\lambda\to\infty} \boldsymbol{\varphi}(s, x) \qquad \text{for } \mathsf{Q}_\Phi\text{-almost every } (s, x), \tag{5.13}$$

in the product topology, then $\max_{1\leq j\leq d} \varphi_{j,S}(X) = \Phi_S(X) < \infty$ almost surely and, under the same conditioning,

$$\big(\zeta_\tau\big(G_j(RU) - \tau^\gamma\big)\big)_{j=1}^d \xrightarrow[\tau\to\infty]{d} (E + \Phi_S(X) - \varphi_{j,S}(X))_{j=1}^d \tag{5.14}$$

in $[0, \infty]^d$. If the maximiser J_* of $(\varphi_{j,S}(X))_{j=1}^d$ is almost surely unique, then conditionally on $E_\tau(\mathbf{1}_d)$

$$(\zeta_\tau(M_\tau - \tau^\gamma), J_\tau) \xrightarrow[\tau\to\infty]{d} (E, J_*), \qquad E \text{ independent of } J_*, \tag{5.15}$$

where

$$\mathbb{P}\{J_* = j\} = \frac{1}{C_\Phi} \int_{\mathcal{M}} h(s) \int_{N_s\mathcal{M}} \mathbf{1}\left\{\varphi_{j,s}(x) > \max_{1\leq i\leq d,\ i\neq j} \varphi_{i,s}(x)\right\} e^{-\Phi_s(x)} \mathrm{d}x \mathrm{d}V_{\mathcal{M}}(s). \tag{5.16}$$

Remark 5.3. *The minimum limit (5.11) holds without (5.13). For the individual excesses, almost-everywhere branch convergence suffices. Ties of positive Q_Φ-measure require a separate argument for the minimising index. For arbitrary positive $\mathbf{b}$ apply Corollary 5.2 to G_j/b_j, $1 \leq j \leq d$.*

Corollary 5.4. *Under Theorem 5.1, if $\overline{F}(r) \sim K r^a e^{-Lr^\beta}$ as $r \to \infty$ with $K, L, \beta > 0$ and $a \in \mathbb{R}$, then $w(r) = L\beta r^{\beta-1}$ is admissible and*

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau\to\infty}{\sim} K C_\Phi (L\beta)^{-\kappa} r_\tau^{a-\beta\kappa} e^{-Lr_\tau^\beta}. \tag{5.17}$$

Let $Z \sim N_k(0, I_k)$ and let G be continuous and positively homogeneous of degree $\gamma > 0$. Suppose $g = G|_{\mathbb{S}^{k-1}}$ attains $M = \max_{s\in\mathbb{S}^{k-1}} g(s) > 0$ precisely on a compact boundaryless C^2 manifold $\mathcal{M}$ of dimension m and codimension $c = k - 1 - m \geq 1$. If, uniformly in normal coordinates,

$$g(\Psi(s, z)) = M - \frac{1}{2}\langle B_s z, z\rangle + o(\|z\|^2), \qquad z \to 0,$$

for a continuous field of symmetric positive-definite operators B_s on $N_s\mathcal{M}$, then with $r_u = (u/M)^{1/\gamma}$

$$\mathbb{P}\{G(Z) > u\} \underset{u\to\infty}{\sim} \frac{(\gamma M)^{c/2}}{(2\pi)^{(m+1)/2}} r_u^{m-1} e^{-r_u^2/2} \int_{\mathcal{M}} \frac{\mathrm{d}V_{\mathcal{M}}(s)}{\sqrt{\det B_s}}. \tag{5.18}$$

Here $\det B_s$ is computed in an orthonormal normal frame.

Formula (5.18) recovers the leading Gaussian-chaos term in [13, Corollary 7] and [9, Theorem 1.2].

5.1. **Weighted angular laws and moving thresholds.** Let μ be an arbitrary angular probability measure. For the moving event (2.6) let ρ_τ be its required radius in (2.5). Fix a tubular neighbourhood $\mathcal{T}$ of a compact boundaryless C^2 manifold $\mathcal{M}$ with coordinates $u = \Psi(s,z)$. Choose $r_\tau \to \infty$ such that

$$\lambda_\tau = r_\tau w(r_\tau) \xrightarrow{\tau\to\infty} \infty. \tag{5.19}$$

Let $D_{\lambda,s} : N_s\mathcal{M} \to N_s\mathcal{M}$ be invertible and continuous in s with

$$\sup_{s\in\mathcal{M}} \|D_{\lambda,s}\| \xrightarrow{\lambda\to\infty} 0. \tag{5.20}$$

For $u = \Psi(s,z) \in \mathcal{T}$ put

$$\Xi_\tau(u) = \big(s, D^{-1}_{\lambda_\tau,s} z\big) \in \mathsf{N}\mathcal{M}, \tag{5.21}$$

and

$$\phi_\tau(u) = w(r_\tau)\big(\rho_\tau(u) - r_\tau\big). \tag{5.22}$$

We shall assume $\phi_\tau \ge -A$ on $\mathcal{T}$ for some fixed $A < \infty$. On the compactified interval $\mathbb{I}_A = [-A,\infty]$ set $e^{-\infty} = 0$. For $a_\tau > 0$ deterministic define the rescaled measure on $\mathsf{N}\mathcal{M}\times\mathbb{I}_A$ by

$$\mathfrak{G}_\tau(B) = \frac{1}{a_\tau}\int_{\mathcal{T}} \mathbf{1}\big\{\big(\Xi_\tau(u),\phi_\tau(u)\big)\in B\big\}\,\mu(\mathrm{d}u). \tag{5.23}$$

By (2.12) and local uniformity on $[-A,0]$, for every $p>0$ and all sufficiently large τ

$$\frac{\overline{F}(r_\tau + t/w(r_\tau))}{\overline{F}(r_\tau)} \le C_A(1+t_+)^{-p}, \qquad t \ge -A, \tag{5.24}$$

where $t_+ = \max(t,0)$.

Theorem 5.5. *Under (2.9) and (5.20) suppose that*

$$\mathfrak{G}_\tau \xrightarrow[\tau\to\infty]{v} \mathfrak{G} \tag{5.25}$$

on $\mathsf{N}\mathcal{M}\times\mathbb{I}_A$ for a Radon measure $\mathfrak{G}$. For some $p>0$ and continuous $\varrho : \mathsf{N}\mathcal{M} \to [0,\infty)$ with compact sublevel sets assume

$$\lim_{R\to\infty}\limsup_{\tau\to\infty}\int_{\{\varrho(\xi)>R\}\times\mathbb{I}_A} (1+t_+)^{-p}\mathfrak{G}_\tau(\mathrm{d}\xi,\mathrm{d}t) = 0, \tag{5.26}$$

and

$$\int_{\mathcal{S}\setminus\mathcal{T}} \overline{F}\big(\rho_\tau(u)\big)\mu(\mathrm{d}u) = o\big(a_\tau\overline{F}(r_\tau)\big), \qquad \tau\to\infty. \tag{5.27}$$

If

$$C_{\mathfrak{G}} := \int_{\mathsf{N}\mathcal{M}\times\mathbb{I}_A} e^{-t}\mathfrak{G}(\mathrm{d}\xi,\mathrm{d}t) > 0, \tag{5.28}$$

then $C_{\mathfrak{G}} < \infty$ and

$$\mathbb{P}\{E_\tau(\mathbf{t}_\tau)\} \underset{\tau\to\infty}{\sim} a_\tau\overline{F}(r_\tau)C_{\mathfrak{G}}. \tag{5.29}$$

Conditionally on $E_\tau(\mathbf{t}_\tau)$ we have

$$\big(\Xi_\tau(U),\phi_\tau(U)\big) \xrightarrow[\tau\to\infty]{d} \mathsf{Q}_{\mathfrak{G}}, \qquad \mathsf{Q}_{\mathfrak{G}}(\mathrm{d}\xi,\mathrm{d}t) = C_{\mathfrak{G}}^{-1}e^{-t}\mathfrak{G}(\mathrm{d}\xi,\mathrm{d}t), \tag{5.30}$$

with arbitrary extensions off $\mathcal{T}$, and jointly

$$w(r_\tau)\big(R-\rho_\tau(U)\big) \;\Big|\; E_\tau(\mathbf{t}_\tau) \xrightarrow[\tau\to\infty]{d} \mathrm{Exp}(1), \tag{5.31}$$

with the exponential limit independent of the pair in (5.30).

For fixed costs Theorem 3.1 derives polynomial tightness from regular variation of the angular Laplace mass. For moving thresholds Theorem 5.5 uses (5.25)–(5.27), with polynomial domination of the radial tail ratio by (5.24). Direct application of the profile extension in [10, Theorem 3.12] would require exponential domination. Uniform angular separation and (2.12) imply (5.27) when their bound for the complement divided by $\overline{F}(r_\tau)$ is $o(a_\tau)$. For moving thresholds this estimate must hold uniformly.

In Theorem 5.1 take $A = 0$, $a_\tau = \lambda_\tau^{-\kappa}$ and $\phi_\tau = \lambda_\tau(f-1)$. The limiting graph measure is

$$h(s)\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x\delta_{\Phi_s(x)}(\mathrm{d}t).$$

The integrable bound and angular separation there imply (5.26) and (5.27). For product-power densities and the Dirichlet application see Sections A and B.

5.2. **Two finite-output normal forms.** We return to the fixed thresholds and smooth setting of Theorem 5.1, with $\mathcal{M}$ compact and boundaryless and $c = n_{\mathcal{S}} - m \geq 1$. We shall assume (2.9), (2.11) and (2.12) with angular density h continuous and positive near $\mathcal{M}$. In both propositions the active sets are fixed along $\mathcal{M}$, the coefficients are continuous in s, and the remainders and local slackness are uniform in s.

Proposition 5.6. *Suppose one active cost satisfies*

$$f_0(\Psi(s,z)) = 1 + \frac{1}{2}\langle H_s z, z\rangle + o(\|z\|^2), \qquad z \to 0, \tag{5.32}$$

with H_s positive definite and the other active costs satisfy

$$f_j(\Psi(s,z)) - f_0(\Psi(s,z)) = L_{j,s}(z) + O(\|z\|^2), \qquad j \in \mathcal{W}, \quad z \to 0, \tag{5.33}$$

with $L_{j,s}$ nonzero linear forms. All remaining costs are locally slack. Let

$$\mathcal{C}_s = \{x \in N_s\mathcal{M} : L_{j,s}(x) \leq 0 \text{ for every } j \in \mathcal{W}\} \tag{5.34}$$

have positive Gaussian measure. For $D_{\lambda,s} = \lambda^{-1/2} I$ the phase and exponent are

$$\Phi_s(x) = \begin{cases} \frac{1}{2}\langle H_s x, x\rangle, & x \in \mathcal{C}_s, \\ \infty, & x \notin \mathcal{C}_s, \end{cases} \qquad \kappa = \frac{c}{2}, \tag{5.35}$$

up to arbitrary values on $\bigcup_{j\in\mathcal{W}}\{L_{j,s} = 0\}$. For $Z_s \sim N(0, H_s^{-1})$ on $N_s\mathcal{M}$ we have

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau\to\infty}{\sim} \overline{F}(r_\tau)\lambda_\tau^{-c/2}(2\pi)^{c/2}\int_{\mathcal{M}} \frac{h(s)\mathbb{P}\{Z_s \in \mathcal{C}_s\}}{\sqrt{\det H_s}}\mathrm{d}V_{\mathcal{M}}(s). \tag{5.36}$$

Proposition 5.7. *Let $N_s\mathcal{M} = \mathsf{V}_s \oplus \mathsf{W}_s$ be a continuous orthogonal splitting with $\dim \mathsf{V}_s = r$ and $\dim \mathsf{W}_s = c - r$. For $y \in \mathsf{V}_s$ and $w \in \mathsf{W}_s$ suppose*

$$f_j(\Psi(s, y+w)) = 1 + \ell_{j,s}(y) + \frac{1}{2}\langle H_{j,s} w, w\rangle + o(\|y\| + \|w\|^2), \qquad j \in \mathcal{A}, \tag{5.37}$$

uniformly as $(y,w) \to (0,0)$, with all other costs locally slack. If

$$\Phi_s(x,z) = \max_{j\in\mathcal{A}}\left(\ell_{j,s}(x) + \frac{1}{2}\langle H_{j,s} z, z\rangle\right) \tag{5.38}$$

satisfies $\Phi_s(x,z) \geq c_1(\|x\| + \|z\|^2)$ for some $c_1 > 0$ independent of s, then in a sufficiently small tube there is $c_0 > 0$ with

$$f(\Psi(s, y+w)) - 1 \geq c_0(\|y\| + \|w\|^2). \tag{5.39}$$

For $D_{\lambda,s}(x+z) = \lambda^{-1}x + \lambda^{-1/2}z$ we have

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau\to\infty}{\sim} \overline{F}(r_\tau)\lambda_\tau^{-(c+r)/2}\int_{\mathcal{M}} h(s)\int_{\mathsf{V}_s\times\mathsf{W}_s} e^{-\Phi_s(x,z)}\mathrm{d}x\mathrm{d}z\mathrm{d}V_{\mathcal{M}}(s). \tag{5.40}$$

In (5.36) the weak constraints contribute a single Gaussian cone probability. The power in (5.40) depends on the switching rank r. If additional weak branches restrict z to a fixed full-dimensional cone, Theorem 5.1 applies with phase ∞ outside that cone once its hypotheses have been verified.

Remark 5.8. *For a fixed active set let*

$$P_s = \operatorname{conv}\{\nabla_N f_j(s) : f_j(s) = 1\}.$$

Local minimality of f implies $0 \in P_s$. The minimal face containing zero spans the switching space. Its support function is coercive on that span since zero lies in its relative interior. The projections of the remaining gradients onto the orthogonal complement determine the weak inequalities, and an exposing normal yields an interior cone direction. The rank and combinatorial type must be constant along $\mathcal{M}$ for the above normal forms. A change of active set or rank, loss of cone interior or intersection of minimising strata requires a separate local analysis. Only after verifying uniform integrability on each stratum can one compare powers. Borderline cases can introduce logarithmic factors.

6. Explicit two-output normal forms

Suppose the minimising set $\mathcal{M}$ in (2.8) is a compact m-dimensional C^2 submanifold of $\mathcal{S}$ without boundary and put $c = n_{\mathcal{S}} - m \geq 1$. In a fixed tube $\Psi(s,z)$, $z \in N_s\mathcal{M}$, with volume Jacobian one at $z = 0$, assume that $f_i = q_i/q_*$, $i = 1, 2$, are C^2 and that μ has a continuous positive density h with respect to Riemannian volume. Write $f = q/q_* = f_1 \vee f_2$ and $h(s) = h(\Psi(s,0))$. The coefficient fields below are continuous and the remainders and lower bounds are uniform on $\mathcal{M}$. Determinants are taken in orthonormal frames.

We distinguish the following local conditions.

(S) For a symmetric positive-definite operator H_s on $N_s\mathcal{M}$,

$$f_1(\Psi(s,z)) = 1 + \tfrac{1}{2}\langle H_s z, z\rangle + o(\|z\|^2), \qquad z \to 0, \tag{6.1}$$

and $f_2 \leq 1 - \eta$ near $\mathcal{M}$ for some $\eta > 0$.

(W) Expansion (6.1) holds with H_s as in (S) and for a nonzero linear form L_s on $N_s\mathcal{M}$

$$f_2(\Psi(s,z)) - f_1(\Psi(s,z)) = L_s(z) + O(\|z\|^2), \qquad z \to 0. \tag{6.2}$$

(A) For a continuous unit normal field e_s put $\mathsf{W}_s = N_s\mathcal{M} \cap e_s^{\perp}$. For $z \in \mathsf{W}_s$ and $a_s, \delta_s > 0$ assume that as $(y,z) \to (0,0)$

$$f_1(\Psi(s, ye_s + z)) = 1 + a_s y + \tfrac{1}{2}\langle H_{1,s} z, z\rangle + o(|y| + \|z\|^2), \tag{6.3}$$

$$f_2(\Psi(s, ye_s + z)) = 1 - \delta_s y + \tfrac{1}{2}\langle H_{2,s} z, z\rangle + o(|y| + \|z\|^2). \tag{6.4}$$

The reduced Hessian

$$\overline{H}_s = \frac{\delta_s H_{1,s} + a_s H_{2,s}}{a_s + \delta_s} \tag{6.5}$$

is uniformly positive definite on W_s.

Under (S) or (W) put

$$C_{\mathrm{S}} = (2\pi)^{c/2} \int_{\mathcal{M}} \frac{h(s)}{\sqrt{\det H_s}} \mathrm{d}V_{\mathcal{M}}(s), \qquad C_{\mathrm{W}} = \tfrac{1}{2} C_{\mathrm{S}}. \tag{6.6}$$

Under (A) put

$$C_{\mathrm{A}} = (2\pi)^{(c-1)/2} \int_{\mathcal{M}} \frac{h(s)(a_s^{-1} + \delta_s^{-1})}{\sqrt{\det \overline{H}_s}} \mathrm{d}V_{\mathcal{M}}(s), \tag{6.7}$$

with $\det \overline{H}_s = 1$ if $c = 1$.

Theorem 6.1. *Assume (2.9) and write $P_\tau = \mathbb{P}\{E_\tau(b)\}/\overline{F}(r_\tau)$. As $\tau \to \infty$*

$$P_\tau \sim C_{\mathrm{S}} \lambda_\tau^{-c/2} \qquad \text{under (S)}, \tag{6.8}$$

$$P_\tau \sim C_{\mathrm{W}} \lambda_\tau^{-c/2} \qquad \text{under (W)}, \tag{6.9}$$

$$P_\tau \sim C_{\mathrm{A}} \lambda_\tau^{-(c+1)/2} \qquad \text{under (A)}. \tag{6.10}$$

In case (A) there is $c_0 > 0$ such that in a sufficiently small tube

$$f(\Psi(s, ye_s + z)) - 1 \geq c_0(|y| + \|z\|^2). \tag{6.11}$$

Remark 6.2. *In (S) and (W) the branches may be interchanged. In (A) the normal gradients are antiparallel by (6.3)–(6.4) and $\nabla_N f_1(s) = a_s e_s$ orients the switching line. For disjoint minimising manifolds of equal radial cost only those with the smallest κ contribute to the leading term and their constants add. If both branches tie with zero normal gradients then the phase is*

$$\frac{1}{2}\max(\langle H_{1,s} z, z\rangle, \langle H_{2,s} z, z\rangle),$$

and Theorem 5.1 applies if this phase is uniformly coercive.

7. Centred Gaussian processes on a fixed finite grid

Let $\mathbf{Z}$ be centred Gaussian in $\mathbb{R}^N$ with covariance $\Gamma = A_\Gamma A_\Gamma^\top$ of rank $r \geq 1$, where $A_\Gamma \in \mathbb{R}^{N\times r}$. Fix symmetric positive semidefinite $N \times N$ matrices $M_1, \dots, M_d$ and $\mathbf{b} \in (0,\infty)^d$.

Proposition 7.1. *Define the quadratic forms*

$$Y_j = \mathbf{Z}^\top M_j \mathbf{Z}, \qquad j = 1, \dots, d. \tag{7.1}$$

With

$$Q_j = A_\Gamma^\top M_j A_\Gamma, \qquad g_j(v) = v^\top Q_j v, \qquad v \in \mathbb{S}^{r-1}, \tag{7.2}$$

and

$$q(v) = \max_{1\leq j\leq d} \left(\frac{b_j}{g_j(v)}\right)^{1/2}, \tag{7.3}$$

where $b_j/0 = \infty$, the event below has probability zero if some $Q_j = 0$, since then $Y_j = 0$ almost surely. Otherwise the minimum $q_* = \min_{v\in\mathbb{S}^{r-1}} q(v)$ lies in $(0,\infty)$ and for every $u > 0$

$$\mathbb{P}\{Y_j > b_j u,\ j = 1, \dots, d\} = \int_{\mathbb{S}^{r-1}} \overline{F}_{\chi_r}\big(\sqrt{u} q(v)\big)\sigma_{r-1}(\mathrm{d}v), \tag{7.4}$$

where σ_{r-1} is normalised surface measure and $\overline{F}_{\chi_r}(x) = \mathbb{P}\{\chi_r > x\}$. Under the respective angular hypotheses the scales are

$$\tau = \sqrt{u}, \qquad r_\tau = \tau q_*, \qquad w(x) = x, \qquad \lambda_\tau = r_\tau w(r_\tau) = u q_*^2. \tag{7.5}$$

Any other full-rank factor has the form $A_\Gamma O$ with O orthogonal, so the angular forms and their minimisers change only by rotation or reflection.

For observations on a finite grid, the squared norm of a block $P_j\mathbf{Z}$ corresponds to $M_j = P_j^\top P_j$ in (7.1). For ν independent Brownian components the (i,j) covariance block is $\min(t_i, t_j) I_\nu$.

Now let $X_1, \dots, X_\nu$ be independent copies of a centred GP on the fixed grid $T = \{t_1, \dots, t_d\}$, $d \geq 2$, $\nu \geq 1$. Put

$$K_{ij} = \mathbb{E}\{X_1(t_i) X_1(t_j)\}, \qquad Y_j := \sum_{\ell=1}^{\nu} X_\ell^2(t_j),$$

and assume $K_{jj} > 0$ for $1 \leq j \leq d$. Write $K = BB^\top$ with $B \in \mathbb{R}^{d\times r}$, $r = \operatorname{rank} K$, and denote row j of B by $b_j^\top$. We study the event, minimum and first minimising index

$$A_u := \{Y_j > u,\ 1 \leq j \leq d\}, \qquad M_u := \min_{1\leq j\leq d} Y_j, \qquad J_u := \min \operatorname*{argmin}_{1\leq j\leq d} Y_j.$$

Corollary 7.2. *For the forms*

$$Q_j = I_\nu \otimes (b_j b_j^\top), \qquad g_j(v) = v^\top Q_j v, \qquad v \in \mathbb{S}^{\nu r-1}, \tag{7.6}$$

put

$$\underline{g}(v) := \min_{1\le j\le d} g_j(v), \qquad g_* := \max_{v\in\mathbb{S}^{\nu r-1}} \underline{g}(v), \qquad q_* := g_*^{-1/2}. \tag{7.7}$$

Then $g_* > 0$. Under the hypotheses of Theorem 5.1 with exponent κ and phase Φ we have

$$\mathbb{P}\{A_u\} \underset{u\to\infty}{\sim} C_\Phi \left(\frac{u}{g_*}\right)^{-\kappa} \overline{F}_{\chi_{\nu r}}\left(\sqrt{\frac{u}{g_*}}\right) \tag{7.8}$$

$$\underset{u\to\infty}{\sim} \frac{2^{1-\nu r/2} C_\Phi}{\Gamma(\nu r/2)} \left(\frac{u}{g_*}\right)^{\nu r/2-1-\kappa} \exp\left(-\frac{u}{2g_*}\right). \tag{7.9}$$

With $E \sim \mathrm{Exp}(1)$, conditionally on A_u

$$M_u - u \xrightarrow[u\to\infty]{d} 2g_* E. \tag{7.10}$$

If further the individual phases φ_j satisfy the branchwise convergence of Corollary 5.2, take $(S, X) \sim \mathsf{Q}_\Phi$ independent of E. Conditionally on A_u

$$(Y_j - u)_{j=1}^d \xrightarrow[u\to\infty]{d} 2g_* \left(E + \Phi_S(X) - \varphi_{j,S}(X)\right)_{j=1}^d \tag{7.11}$$

in $[0,\infty]^d$. If the maximising phase is Q_Φ-almost surely unique then

$$(M_u - u, J_u) \mid A_u \xrightarrow[u\to\infty]{d} (2g_* E, J_*), \qquad J_* = \operatorname*{argmax}_{1\le j\le d} \varphi_{j,S}(X), \tag{7.12}$$

with E independent of J_*. The laws depend only on K and ν.

Remark 7.3. *For $K = aa^\top$ with all $a_j \neq 0$ write $Y_j = a_j^2 V$ jointly, where $V \sim \chi^2_\nu$. With $m_0 = \min_{1\le j\le d} a_j^2$ and $j_0 = \min \operatorname{argmin}_{1\le j\le d} a_j^2$ we have*

$$A_u = \{V > u/m_0\}, \qquad M_u = m_0 V, \qquad J_u = j_0.$$

Thus $\mathbb{P}\{A_u\} = \mathbb{P}\{\chi^2_\nu > u/m_0\}$ and

$$(M_u - u, J_u) \mid A_u \xrightarrow[u\to\infty]{d} (2m_0 E, j_0), \qquad E \sim \mathrm{Exp}(1).$$

Here every angle minimises the cost.

For the next result assume K positive definite. After relabelling if necessary put

$$\sigma^2 := K_{11}, \qquad \rho_j := \frac{K_{j1}}{\sigma^2}, \quad 2 \le j \le d.$$

Suppose that the regression coefficients satisfy $\rho_j^2 \ge 1$ for $2 \le j \le d$ and put

$$\mathcal{W} := \{j \ge 2 : \rho_j^2 = 1\}, \qquad \mathcal{S}_0 := \{j \ge 2 : \rho_j^2 > 1\}.$$

Corollary 7.4. *Let $\mathbf{Z}^\circ = (Z_2^\circ, \dots, Z_d^\circ) \sim N(0, K^\circ)$ with*

$$K^\circ_{ij} := K_{ij} - \frac{K_{i1}K_{j1}}{\sigma^2}, \qquad 2 \le i, j \le d, \tag{7.13}$$

and put

$$\pi_0 := \mathbb{P}\{\rho_j Z_j^\circ > 0,\ j \in \mathcal{W}\}, \tag{7.14}$$

with $\pi_0 = 1$ for $\mathcal{W} = \emptyset$. The Schur complement K° is positive definite, hence $\pi_0 > 0$. Write $\mathbf{Z}^{\circ,+}$ for $\mathbf{Z}^\circ$ conditioned on the event in (7.14). Then

$$\mathbb{P}\{A_u\} \underset{u\to\infty}{\sim} \pi_0 \mathbb{P}\{\sigma^2 \chi^2_\nu > u\} \tag{7.15}$$

$$\underset{u\to\infty}{\sim} \frac{\pi_0}{\Gamma(\nu/2)} \left(\frac{u}{2\sigma^2}\right)^{\nu/2-1} \exp\left(-\frac{u}{2\sigma^2}\right). \tag{7.16}$$

Conditionally on A_u

$$\left(Y_1-u,\ \left(\frac{Y_j-u}{2\sqrt{u}}\right)_{j\in\mathcal{W}},\ \left(\frac{Y_j-u}{u}\right)_{j\in\mathcal{S}_0}\right)\xrightarrow[u\to\infty]{d}\left(E_0,\ (\rho_j Z_j^{\circ,+})_{j\in\mathcal{W}},\ (\rho_j^2-1)_{j\in\mathcal{S}_0}\right), \tag{7.17}$$

where $E_0\sim\mathrm{Exp}(1/(2\sigma^2))$ *is independent of* $\mathbf{Z}^{\circ,+}$*, with empty subvectors omitted. In particular*

$$(M_u-u,J_u)\ \big|\ A_u\xrightarrow[u\to\infty]{d}(E_0,1). \tag{7.18}$$

Remark 7.5. *For* $K=\mathrm{diag}(\sigma_1^2,\dots,\sigma_d^2)$ *the* Y_j *are independent with laws* $\sigma_j^2\chi^2_\nu$*. Then*

$$g_*=\left(\sum_{j=1}^d\sigma_j^{-2}\right)^{-1},$$

and conditionally on A_u *the excesses converge to independent exponentials of rates* $1/(2\sigma_j^2)$*. Hence*

$$\mathbb{P}\{J_u=j\mid A_u\}\xrightarrow{u\to\infty}\frac{\sigma_j^{-2}}{\sum_{i=1}^d\sigma_i^{-2}}.$$

The limiting minimum excess has rate $\frac12\sum_{i=1}^d\sigma_i^{-2}$ *and is independent of the limiting index, which is uniform for equal variances.*

Remark 7.6. *If some* $\rho_j^2<1$ *then* $g_*<\sigma^2$*, though another grid point may satisfy the condition of Corollary 7.4. If none does, several constraints enter at leading order. A singular residual covariance may also produce ties or a lower-dimensional cone. In either case Corollary 7.2 applies whenever its normal-form hypotheses hold.*

Remark 7.7. *To allow the regression coefficients in Corollary 7.4 to vary with* u*, take* $K^{(u)}\to K^{(0)}>0$ *as* $u\to\infty$*. Form* $Y_{j,u}$ *from* ν *independent centred Gaussian copies with covariance* $K^{(u)}$ *as above and put* $A_u=\bigcap_{j=1}^d\{Y_{j,u}>u\}$*. Write*

$$\sigma_u^2=K_{11}^{(u)},\qquad\rho_{j,u}=\frac{K_{j1}^{(u)}}{\sigma_u^2},\qquad K_{u,ij}^{\circ}=K_{ij}^{(u)}-\frac{K_{i1}^{(u)}K_{j1}^{(u)}}{\sigma_u^2},\quad i,j\ge2,$$

and define K_0° *likewise. Suppose for* $j\in\mathcal{T}_0\subset\{2,\dots,d\}$ *that*

$$\rho_{j,u}\xrightarrow{u\to\infty}s_j\in\{-1,1\},\qquad\beta_{j,u}:=(\rho_{j,u}^2-1)\sqrt{u}\xrightarrow{u\to\infty}\beta_j\in\mathbb{R},$$

while

$$\liminf_{u\to\infty}\min_{\substack{j\ge2\\ j\notin\mathcal{T}_0}}(\rho_{j,u}^2-1)>0$$

if the complement is nonempty. The complementary coordinates are uniformly slack and Gaussian regression yields, conditionally on $Y_{1,u}>u$

$$\left(\frac{Y_{j,u}-u}{2\sqrt{u}}\right)_{j\in\mathcal{T}_0}\xrightarrow[u\to\infty]{d}\left(\frac{\beta_j}{2}+s_jZ_j^\circ\right)_{j\in\mathcal{T}_0},\qquad(Z_j^\circ)_{j\in\mathcal{T}_0}\sim N\left(0,(K_0^\circ)_{\mathcal{T}_0\mathcal{T}_0}\right).$$

Consequently

$$\frac{\mathbb{P}\{A_u\}}{\mathbb{P}\{Y_{1,u}>u\}}\xrightarrow{u\to\infty}\mathbb{P}\left\{\frac{\beta_j}{2}+s_jZ_j^\circ>0,\ j\in\mathcal{T}_0\right\}.$$

For a single transition coordinate the factor is $\Phi_{\mathrm{N}}(\beta_j/(2\sqrt{K_{0,jj}^\circ}))$*, with* Φ_{N} *the standard normal distribution function. Thus* $(\rho_{j,u}^2-1)\sqrt{u}=O(1)$ *is the transition scale, with factor* $1/2$ *at* $\beta_j=0$ *and limit* 1 *when* $\beta_{j,u}\to+\infty$*. With several zero shifts the factor is the joint Gaussian orthant probability.*

Remark 7.8. *The uniform angle in Proposition 7.1 has the symmetrised Dirichlet law with all $\alpha_i = 1/2$. For the degree-one orthant outputs of Corollary B.3 these parameters and $R \sim \chi_k$ recover the Gaussian case. A general Gumbel radius is allowed there, and when some $\alpha_i \neq 1/2$ the factors in (B.1) depend on more than the covariance.*

7.1. **Brownian motion on a fixed grid.** For standard Brownian motion $B = (B_1, \dots, B_\nu)$ in $\mathbb{R}^\nu$ with $\nu \geq 1$ put

$$Y_\nu(t) = \|B(t)\|^2 = \sum_{\ell=1}^{\nu} B_\ell^2(t). \tag{7.19}$$

For $\mathbf{T} = (t_1, \dots, t_d)$ with $d \geq 2$ and $0 < t_1 < \cdots < t_d$ set

$$A_u^{(d)} = \bigcap_{j=1}^{d} \{Y_\nu(t_j) > u\}. \tag{7.20}$$

If W is a scalar standard Brownian motion, write

$$\pi_d(\mathbf{T}) = \mathbb{P}\{W(t_j - t_1) > 0,\ j = 2, \dots, d\}. \tag{7.21}$$

The Gaussian vector in (7.21) is nondegenerate. For two times and unequal thresholds see Section C.

Corollary 7.9. *As $u \to \infty$*

$$\mathbb{P}\left\{A_u^{(d)}\right\} \sim \pi_d(\mathbf{T}) \mathbb{P}\{Y_\nu(t_1) > u\} \tag{7.22}$$

$$\sim \frac{\pi_d(\mathbf{T})}{\Gamma(\nu/2)} \left(\frac{u}{2t_1}\right)^{\nu/2-1} \exp\left(-\frac{u}{2t_1}\right). \tag{7.23}$$

Conditionally on $A_u^{(d)}$

$$\left(Y_\nu(t_1) - u, \left(\frac{Y_\nu(t_j) - u}{2\sqrt{u}}\right)_{j=2}^{d}\right) \xrightarrow[u\to\infty]{d} \left(E, \big(W(t_j - t_1)\big)_{j=2}^{d} \;\middle|\; W(t_j - t_1) > 0,\ j = 2, \dots, d\right), \tag{7.24}$$

where $E \sim \mathrm{Exp}(1/(2t_1))$ is independent of the conditional Gaussian vector. In particular

$$\min_{1\leq j\leq d} Y_\nu(t_j) - u \;\Big|\; A_u^{(d)} \xrightarrow[u\to\infty]{d} E, \tag{7.25}$$

and the conditional probability that t_1 attains the minimum tends to one.

7.1.1. *The explicit three-output constant.* For $d = 3$ let $\Delta_i = t_{i+1} - t_i$ for $i = 1, 2$. The variables $W(\Delta_1)$ and $W(\Delta_1 + \Delta_2)$ have correlation $\sqrt{\Delta_1/(\Delta_1 + \Delta_2)}$, whence [15, Section 2.2] yields

$$\pi_3(\mathbf{T}) = \frac{1}{4} + \frac{1}{2\pi} \arcsin \sqrt{\frac{\Delta_1}{\Delta_1 + \Delta_2}} = \frac{1}{4} + \frac{1}{2\pi} \arctan \sqrt{\frac{\Delta_1}{\Delta_2}}. \tag{7.26}$$

For equally spaced times $\pi_3(\mathbf{T}) = 3/8$. The two later coordinates in (7.24) have limiting density

$$\frac{\mathbf{1}\{x > 0, y > 0\}}{2\pi\pi_3(\mathbf{T})\sqrt{\Delta_1\Delta_2}} \exp\left(-\frac{x^2}{2\Delta_1} - \frac{(y-x)^2}{2\Delta_2}\right). \tag{7.27}$$

For $\nu = 2$ formula (7.23) reduces to

$$\mathbb{P}\left\{\min_{1\leq j\leq 3} Y_2(t_j) > u\right\} \underset{u\to\infty}{\sim} \pi_3(\mathbf{T}) \exp\left(-\frac{u}{2t_1}\right). \tag{7.28}$$

Remark 7.10. *When grid times coalesce the limiting covariance in (7.21) is singular and (7.22) need not hold uniformly. A dense grid leads to a Brownian persistence problem and requires process localisation.*

8. A uniform growing-dimension radial benchmark

Let $Z^{(k)} \sim N_k(0, I_k)$ and $H_k(z) = \sum_{i=1}^k z_i^2$ for $z \in \mathbb{R}^k$. Since $H_k = 1$ on the unit sphere the angular set has codimension zero. For integer $k \geq 1$ and $x > 0$ set

$$a = k/2, \qquad p = a - 1, \qquad p_+ = \max(p, 0), \qquad L_k(x) = x^{a-1}e^{-x}/\Gamma(a).$$

Proposition 8.1. *For $x > 0$*

$$\mathbb{P}\Big\{H_k(Z^{(k)}) > 2x\Big\} = L_k(x)R(a, x), \qquad R(a, x) = \int_0^\infty e^{-t}(1 + t/x)^p \mathrm{d}t. \tag{8.1}$$

For $x > p_+$

$$|R(a, x) - 1| \leq \frac{|p|}{x - p_+}. \tag{8.2}$$

If $k \geq 2$ and $\beta = 1 - p/x > 0$ then

$$\mathbb{P}\Big\{H_k(Z^{(k)}) > 2x\Big\} = \frac{L_k(x)}{\beta}(1 + \varepsilon), \qquad -\frac{p}{x^2\beta^2} \leq \varepsilon \leq 0. \tag{8.3}$$

Fix $0 < b_0 \leq b_1 < \infty$ and an integer sequence $k = k(u) \geq 1$. If $k = o(u)$ as $u \to \infty$ then uniformly in $b \in [b_0, b_1]$

$$\mathbb{P}\Big\{H_k(Z^{(k)}) > bu\Big\} = \frac{(bu/2)^{k/2-1}e^{-bu/2}}{\Gamma(k/2)}\left(1 + O\left(\frac{k}{u}\right)\right). \tag{8.4}$$

If $k/u \to c \in [0, b_0)$ set $\beta_{u,b} = 1 - (k - 2)/(bu)$. Then uniformly in b

$$\mathbb{P}\Big\{H_k(Z^{(k)}) > bu\Big\} \sim \frac{(bu/2)^{k/2-1}e^{-bu/2}}{\Gamma(k/2)\beta_{u,b}} \sim \frac{b}{b - c}\frac{(bu/2)^{k/2-1}e^{-bu/2}}{\Gamma(k/2)}, \tag{8.5}$$

and

$$\sup_{b\in[b_0,b_1]} \sup_{y\geq 0} \left|\mathbb{P}\left\{\frac{\beta_{u,b}}{2}\big(H_k(Z^{(k)}) - bu\big) > y \middle| H_k(Z^{(k)}) > bu\right\} - e^{-y}\right| \xrightarrow{u\to\infty} 0. \tag{8.6}$$

For fixed b the limiting unscaled excess has exponential rate $(b - c)/(2b)$, equal to $1/2$ when $k = o(u)$.

Proof. Write $S_k(x) = \mathbb{P}\{H_k(Z^{(k)}) > 2x\}$. Substitution $s = x + t$ in the gamma tail yields (8.1). For $p < 0$ convexity yields $1 + pt/x \leq (1 + t/x)^p \leq 1$. For $p \geq 0$ and $x > p$ the bounds $v - v^2/2 \leq \log(1 + v) \leq v$ and $e^{-v} \geq 1 - v$ for $v \geq 0$ yield

$$1 \leq R(a, x) \leq \beta^{-1}, \qquad R(a, x) \geq \int_0^\infty e^{-\beta t}\left(1 - \frac{pt^2}{2x^2}\right)\mathrm{d}t = \beta^{-1} - \frac{p}{x^2\beta^3}.$$

Integration in the first case and these bounds in the second prove (8.2)–(8.3).

Take $x = bu/2$ with $b \in [b_0, b_1]$. If $k = o(u)$ then (8.2) yields (8.4). If $k/u \to c < b_0$ then $\beta_{u,b} - (1 - c/b) \to 0$ uniformly in b and $\inf_{b\in[b_0,b_1]} \beta_{u,b}$ stays positive. Thus (8.3) yields (8.5) with relative error $O(u^{-1})$ in its first equivalent for $k \geq 2$. For $k = 1$ use (8.2).

Since $-S_k'(x) = L_k(x)$, for each fixed $y \geq 0$

$$\frac{S_k(x + y/\beta_{u,b})}{S_k(x)} = \exp\left(-\int_0^{y/\beta_{u,b}} \frac{\mathrm{d}t}{R(a, x + t)}\right) \longrightarrow e^{-y}, \qquad u \to \infty,$$

uniformly in b. Indeed on bounded t-intervals $1 - p/(x + t) = \beta_{u,b} + O(u^{-1})$, so (8.3) for $k \geq 2$ and (8.2) for $k = 1$ yield $1/R(a, x + t) = \beta_{u,b} + O(u^{-1})$ uniformly. Monotonicity of the ratio in y extends convergence from a finite mesh to compact intervals and then to $[0, \infty)$ since $e^{-y} \downarrow 0$. This proves (8.6). □

Keeping $\Gamma(k/2)$ covers bounded k and avoids the $O(1/k)$ relative error of Stirling's formula when $k \to \infty$. The bounds in (8.5)–(8.6) require $c < b_0$.

For a fixed rv a tail equivalent at $v \to \infty$ holds uniformly at $v = bu$ on $b \in [b_0, b_1]$. In growing dimension one also needs uniform control of the angular phase, amplitude and complement relative to the leading term, and of the radial tail over the resulting range of shifts.

For smooth nondegenerate minima, under derivative and localisation bounds, [12, Theorem 2.12 and Corollary 2.17] yields expansions in the small parameter d^2/λ. The logarithmic expansion in [11, Theorem 3.2] has order-L remainder of scale $(R^2 d)^{L+1}/\lambda^L$ under its localisation conditions, hence $O(d^{L+1}/\lambda^L)$ for bounded R. Switching branches and singular angular weights require separate uniform estimates.

9. Proofs

9.1. **Proofs for positive angular measures.**

Proof of Theorem 3.1. Karamata's theorem [1] applied to the Laplace–Stieltjes transform in (3.2) yields (3.3), in particular

$$L_{\mathrm{ang}}(\lambda) \underset{\lambda\to\infty}{\sim} \Gamma(\kappa+1) H(1/\lambda). \tag{9.1}$$

Let ν_λ be the law of $\lambda W(U)$ restricted to $[0,\infty)$ and divided by $H(1/\lambda)$. For each $x > 0$

$$\nu_\lambda([0,x]) = \frac{H(x/\lambda)}{H(1/\lambda)} \xrightarrow{\lambda\to\infty} x^\kappa. \tag{9.2}$$

For $\varepsilon > 0$ Potter's bound applies for $1 \le x \le \delta\lambda$ with $\delta > 0$ sufficiently small. For $x > \delta\lambda$ use $H \le 1$ and $1/H(1/\lambda) = O(\lambda^{\kappa+\varepsilon})$. Thus

$$\nu_\lambda([0,x]) \le C x^{\kappa+\varepsilon}, \qquad x \ge 1, \quad \lambda \ge \lambda_0. \tag{9.3}$$

Summing over $[2^j M, 2^{j+1} M)$, $j \ge 0$, yields for $p > \kappa + \varepsilon$

$$\sup_{\lambda\ge\lambda_0} \int_{[M,\infty)} (1+x)^{-p} \nu_\lambda(\mathrm{d}x) \le C' M^{\kappa+\varepsilon-p}, \qquad M \ge 1. \tag{9.4}$$

Write $K_u(x) = \overline{F}(r_u + x/w(r_u))/\overline{F}(r_u)$. By monotonicity (2.9) yields $K_u(x) \to e^{-x}$ locally uniformly for $x \ge 0$ as $u \to \infty$. For fixed $y \ge 0$ use (9.2) on $[0, M]$ and (2.12), (9.4) on $[M, \infty)$ to obtain

$$\int_{[0,\infty)} |K_u(x+y) - e^{-x-y}| \nu_{\lambda_u}(\mathrm{d}x) \xrightarrow{u\to\infty} 0. \tag{9.5}$$

Put $J_u = \mathbb{P}\{E_u\}/\overline{F}(r_u)$ and $L_u = L_{\mathrm{ang}}(\lambda_u)$. Since $uq(U) = r_u + \lambda_u W(U)/w(r_u)$, conditioning on U yields

$$J_u \mathbb{P}\{U \in A,\ B_u > y \mid E_u\} = \int_{A\cap\{q<\infty\}} K_u(\lambda_u W(v) + y) \mu(\mathrm{d}v).$$

By (9.5) the right-hand side equals $e^{-y} L_u Q_{\lambda_u}(A) + o(H(1/\lambda_u))$ uniformly in $A \in \mathcal{A}$. Taking $A = \mathcal{S}$ and $y = 0$ yields $J_u \sim L_u$ by (9.1), proving (3.5) and (3.8). Division by J_u now yields (3.6) and its special case (3.9).

For a Q-continuity set $D \subset \mathsf{X}$ take $A = T_u^{-1}(D)$ in (3.6). The limiting probability on $D \times (y,\infty)$ is $Q(D)e^{-y}$. The cases $y = 0$ and $D = \mathsf{X}$ yield marginal tightness, and these rectangles identify the joint limit in (3.7). □

Proof of Corollary 3.2. Under Q_{λ_u} the Laplace transform of A_u satisfies

$$\frac{L_{\mathrm{ang}}((1+t)\lambda_u)}{L_{\mathrm{ang}}(\lambda_u)} \xrightarrow{u\to\infty} (1+t)^{-\kappa}, \qquad t \ge 0.$$

The limit is the transform of G_κ, including $G_0 = 0$. The continuity theorem and (3.7) with $T_u = \lambda_u W$ yield (3.10), setting $T_u = 0$ on $\{q = \infty\}$ which is null under both angular laws. On E_u

$$w(r_u)(R - r_u) = A_u + B_u, \qquad q_* w(r_u)\left(\frac{R}{q(U)} - u\right) = \frac{B_u}{1 + A_u/\lambda_u}.$$

Since $G_\kappa + E \sim \mathrm{Gamma}(\kappa + 1, 1)$ and $A_u/\lambda_u \to 0$ in conditional probability as $u \to \infty$, (3.11) and (3.12) follow. The minimum identity follows from (2.1) and (2.3). □

9.2. **Proofs for vanishing angular costs.**

Proof of Theorem 3.5. Fix $0 < \delta < \min(\alpha, \varepsilon)$. By (3.14) and Potter's bound, for some $s_0 > 0$ and all large u

$$\frac{H_0(r/u)}{H_0(1/u)} \le C(r^{\alpha-\delta} + r^{\alpha+\delta}), \qquad 0 < r \le s_0 u. \tag{9.6}$$

For $0 < t \le 1$ and $r > 0$

$$\frac{H_0(tr/u)}{H_0(1/u)} \xrightarrow{u\to\infty} (tr)^\alpha. \tag{9.7}$$

On $\{R \le s_0 u\}$ the bound in (9.6) dominates this ratio at $r = R$ and is integrable by the moment assumption. For bounded measurable g, Markov's inequality yields

$$\frac{\mathbb{E}\{|g(R)|H_0(tR/u)\mathbf{1}_{\{R>s_0u\}}\}}{H_0(1/u)} \le \frac{\|g\|_\infty \mathbb{E}\{R^{\alpha+\varepsilon}\}}{(s_0u)^{\alpha+\varepsilon}H_0(1/u)} \xrightarrow{u\to\infty} 0 \tag{9.8}$$

since $u^\varepsilon \ell_0(u) \to \infty$. Dominated convergence on the first region and (9.8) on its complement yield

$$\frac{\mathbb{E}\{g(R)H_0(tR/u)\}}{H_0(1/u)} \xrightarrow{u\to\infty} t^\alpha \mathbb{E}\{g(R)R^\alpha\}. \tag{9.9}$$

Conditioning on R yields

$$\mathbb{E}\{g(R)\mathbf{1}_{E_u^0}\mathbf{1}_{\{uq(U)/R<t\}}\} = \mathbb{E}\{g(R)H_0(tR/u)\}. \tag{9.10}$$

Taking $g = 1$ and $t = 1$ yields (3.15). Dividing (9.10) by $\mathbb{P}\{E_u^0\}$ and using (3.16) and (9.9) yields

$$\mathbb{E}\{g(R)\mathbf{1}_{\{uq(U)/R<t\}} \mid E_u^0\} \xrightarrow{u\to\infty} t^\alpha \mathbb{E}\{g(R_\alpha)\}. \tag{9.11}$$

Taking $g = \mathbf{1}_{(0,r]}$ identifies the product limit in (3.17) at its continuity points, since t^α is continuous and the ratio lies in $(0, 1)$ on E_u^0. As $B_\alpha > 0$ almost surely, taking its reciprocal yields (3.18). □

9.3. **Proof of the logarithmic angular example.**

Proof of Proposition 4.1. Let $I(z) = \int_D e^{-z\psi(x,y)}\mathrm{d}x\mathrm{d}y$. Integrating first in x and putting $t = z^{1/3}|y|$ as in the collar calculation of [10] yields

$$I(z) = \frac{\sqrt{\pi}}{\sqrt{z}} \int_0^{\varepsilon z^{1/3}} e^{-t^3} \frac{\mathrm{erf}(A_z t)}{t}\mathrm{d}t, \qquad A_z = \varepsilon z^{1/6}, \quad \mathrm{erf}(v) = \frac{2}{\sqrt{\pi}}\int_0^v e^{-s^2}\mathrm{d}s. \tag{9.12}$$

The intervals $t < A_z^{-1}$ and $t > 1$ contribute $O(1)$ to the integral by $\mathrm{erf}(v) \le 2v/\sqrt{\pi}$ and exponential decay. On $[A_z^{-1}, 1]$ replacing the numerator by one costs at most

$$\int_0^1 t^2 \mathrm{d}t + \int_1^\infty (1 - \mathrm{erf}(v))\frac{\mathrm{d}v}{v} < \infty.$$

Thus the integral is $\log A_z + O(1)$. For $D_\delta = (0,\delta)\times(-\delta,\delta)$, $0<\delta<\varepsilon$, the Gaussian bound for $x\ge\delta$ and the cubic bound for $|y|\ge\delta$ yield

$$\int_{D\setminus D_\delta} e^{-z\psi}\mathrm{d}x\mathrm{d}y \le \sqrt{\pi/z}\log(\varepsilon/\delta) + 2\varepsilon^2 e^{-z\delta^3}. \tag{9.13}$$

Since h is bounded, $|L(z)-h_0 I(z)| \le I(z)\sup_{D_\delta}|h-h_0| + O_\delta(z^{-1/2})$. Letting $z\to\infty$ and then $\delta\downarrow 0$ proves (4.6), and (3.5) yields (4.7).

Under the Gibbs law use the coordinates

$$x = z^{-t}, \qquad y = z^{t-1/2}v, \qquad \mathrm{d}x\mathrm{d}y = z^{-1/2}\log z\,\mathrm{d}t\mathrm{d}v. \tag{9.14}$$

The domain and phase become

$$D_z = \{(t,v) : t > \log(1/\varepsilon)/\log z,\ |v| < \varepsilon z^{1/2-t}\}, \tag{9.15}$$

$$z\psi(x,y) = v^2 + z^{3t-1/2}|v|^3. \tag{9.16}$$

On compact subsets of $(0,1/6)\times\mathbb{R}$ we have uniformly

$$h(z^{-t}, z^{t-1/2}v)e^{-v^2 - z^{3t-1/2}|v|^3} \xrightarrow[z\to\infty]{} h_0 e^{-v^2}. \tag{9.17}$$

By (4.6) the limiting Gibbs density on this strip is $6e^{-v^2}/\sqrt{\pi}$. Its integral is one, hence exhaustion by compact rectangles yields the product limit in (4.9) for (T_u, V_u) under the Gibbs law at $z=\lambda_u$. Equation (3.7) adds the independent exponential overshoot under conditioning on E_u. Since $B_u = +$ exactly when $V_u > 0$ off a null set and $\mathbb{P}\{V=0\}=0$, (4.10) follows by continuous mapping. □

9.4. **Abstract normal-form results.**

Proof of Theorem 5.1. Write $D = D_{\lambda_\tau,s}$, $v = \Psi(s,Dx)$ and $A_\tau(s,x) = \lambda_\tau(f(v)-1)$. With J the tubular Jacobian, (2.4) and (5.1) yield

$$\frac{\mathbb{P}\{E_\tau(\mathbf{b}),\ U\in\mathcal{T}\}}{\overline{F}(r_\tau)\lambda_\tau^{-\kappa}} = \int_{\mathsf{N}\mathcal{M}} \frac{\overline{F}(r_\tau + A_\tau(s,x)/w(r_\tau))}{\overline{F}(r_\tau)} h(v)J(s,Dx)\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x, \tag{9.18}$$

with integrand zero off the chart. By (5.2) the domain eventually contains every compact set and $h(v)J(s,Dx)\to h(s)$ there uniformly. For fixed $y\ge 0$ replace A_τ by $A_\tau + y$ in the tail ratio. By (5.3) and local uniformity in (2.9) (monotonicity and a continuous limit), the shifted integrand tends almost everywhere to $e^{-y}h(s)e^{-\Phi_s(x)}$ where $\Phi_s(x)<\infty$. At $\Phi_s(x)=\infty$ it tends to zero by (2.12). After shrinking the tube so that hJ is bounded, (2.12) and (5.4) yield the common bound $C(1+\varrho_s(x))^{-p}$, integrable by (5.5) and compactness. The convergence is therefore in L^1.

Off the tube, (2.11) and (2.12) bound the unshifted contribution by $O(\overline{F}(r_\tau)\lambda_\tau^{-p})$, negligible since $p>\kappa$. At $y=0$ integration yields (5.7) and normalisation of the densities yields (5.8) in total variation. For any bounded continuous angular test function $H(s,x)$ the shifted integral with factor H, divided by the unshifted mass, tends to $e^{-y}\int H\mathrm{d}\mathsf{Q}_\Phi$. The shift imposes $w(r_\tau)(R-\tau q(U)) > y$ and hence proves (5.9) jointly with (5.8) and with independent limits. □

Proof of Corollary 5.4. The Weibull equivalent yields (2.9) with $w(r) = L\beta r^{\beta-1}$. Substitution of $\lambda_\tau = L\beta r_\tau^\beta$ in (5.7) proves (5.17). For Z standard Gaussian the independent radius and angle satisfy

$$\overline{F}_{\chi_k}(r) \underset{r\to\infty}{\sim} \frac{2^{1-k/2}}{\Gamma(k/2)} r^{k-2}e^{-r^2/2}, \qquad h = \frac{\Gamma(k/2)}{2\pi^{k/2}}.$$

Near $\mathcal{M}$ we have uniformly $f = (M/g)^{1/\gamma} = 1 + \langle B_s z, z\rangle/(2\gamma M) + o(\|z\|^2)$. Thus $D_{\lambda,s} = \lambda^{-1/2}I$, $\kappa = c/2$ and the normal integral in (5.6) is $(2\pi\gamma M)^{c/2}/\sqrt{\det B_s}$. Equation (5.7) at $r_u = (u/M)^{1/\gamma}$ yields (5.18) since $k-2-c = m-1$. □

Proof of Corollary 5.2. Condition throughout on $E_\tau(\mathbf{1}_d)$. The fixed tube has conditional probability tending to one. Put $X_\tau = D^{-1}_{\lambda_\tau,S_\tau} z_\tau$, $E^R_\tau = w(r_\tau)(R - \tau q(U))$, $f_{\tau,j} = f_j(U)$ and $f_\tau = f(U)$. By Theorem 5.1

$$(S_\tau, X_\tau, E^R_\tau) \xrightarrow[\tau\to\infty]{d} (S, X, E). \tag{9.19}$$

Tightness of X_τ, (5.2) and continuity of f on a neighbourhood of $\mathcal{M}$ imply $f_\tau \to 1$ in conditional probability. Since $M_\tau = (R/q(U))^\gamma$, Taylor expansion yields

$$\zeta_\tau(M_\tau - \tau^\gamma) = \frac{\lambda_\tau}{\gamma}\left[\left(1 + \frac{E^R_\tau}{\lambda_\tau f_\tau}\right)^\gamma - 1\right] = E^R_\tau + o_{\mathbb{P}}(1), \tag{9.20}$$

proving (5.11).

Suppose now that (5.13) holds. In (9.18) shift A_τ by $y \geq 0$ and insert $H(s, x, \boldsymbol{\varphi}_{\lambda_\tau}(s, x))$ for bounded continuous H. On $\{\Phi < \infty\}$, (5.13) holds bundle-volume almost everywhere since $he^{-\Phi} > 0$ there. On $\{\Phi = \infty\}$ the kernel tends to zero. The domination in (5.4)–(5.5) and normalisation by (5.7) thus yield the limit $e^{-y}\int H(s, x, \boldsymbol{\varphi}(s, x))\mathsf{Q}_\Phi(\mathrm{d}s, \mathrm{d}x)$. Hence (9.19) holds jointly with convergence of the branch phases to $\boldsymbol{\varphi}(S, X)$, whose maximum is $\Phi_S(X) < \infty$ almost surely by (5.3).

By definition

$$R/(\tau q_*) = f_\tau + E^R_\tau/\lambda_\tau. \tag{9.21}$$

Putting $D_{\tau,j} = E^R_\tau + \lambda_\tau(f_\tau - f_{\tau,j})$, homogeneity yields

$$\zeta_\tau(G_j(RU) - \tau^\gamma) = \frac{\lambda_\tau}{\gamma}\left[\left(1 + \frac{D_{\tau,j}}{\lambda_\tau f_{\tau,j}}\right)^\gamma - 1\right]. \tag{9.22}$$

The joint phase limit yields $D_{\tau,j} \Rightarrow E + \Phi_S(X) - \varphi_{j,S}(X)$ jointly over j as $\tau \to \infty$. At a finite limiting phase $f_{\tau,j} \to 1$ and the first-order expansion in (9.22) has this same limit. At phase $-\infty$ we have $D_{\tau,j} \to \infty$ and $0 < f_{\tau,j} \leq f_\tau \to 1$, so the right-hand side eventually exceeds any fixed bound along such convergent sequences. This proves (5.14) by the continuous mapping theorem.

On the tube J_τ is the first maximiser of the branch phases. This map is continuous at a unique finite maximum in the product extended-real topology. Thus (9.20) yields (5.15), since J_* is a function of (S, X) independent of E. Integrating its indicator under Q_Φ proves (5.16). □

Proof of Theorem 5.5. Put $K_\tau(t) = \overline{F}(r_\tau + t/w(r_\tau))/\overline{F}(r_\tau)$ for finite t and $K_\tau(\infty) = 0$. Monotonicity and (2.9) yield uniform convergence to e^{-t} on $[-A, T]$. On $[T, \infty]$ the error is bounded by $K_\tau(T) + e^{-T}$, whence on letting $T \to \infty$

$$\sup_{t\in\mathbb{I}_A} |K_\tau(t) - e^{-t}| \xrightarrow{\tau\to\infty} 0. \tag{9.23}$$

Conditioning on U in (5.23) yields

$$\frac{\mathbb{P}\{E_\tau(\mathbf{t}_\tau),\ U \in \mathcal{T}\}}{a_\tau \overline{F}(r_\tau)} = \int K_\tau(t)\mathfrak{G}_\tau(\mathrm{d}\xi, \mathrm{d}t). \tag{9.24}$$

Let $W(t) = (1 + t_+)^{-p}$ with $W(\infty) = 0$. By (5.25) the measures $W\mathfrak{G}_\tau$ converge vaguely to $W\mathfrak{G}$. The sublevels of ϱ and $\mathbb{I}_A$ are compact, so (5.26) yields tightness, finiteness of the limiting measure and weak convergence. In particular $C_\mathfrak{G} < \infty$ since $e^{-t} \leq CW(t)$.

For fixed $y \geq 0$ and bounded continuous H, (9.23) and vague convergence imply on inserting a continuous compactly supported cutoff in ξ that the integral of $H(\xi, t)K_\tau(t + y)$ converges to that of $H(\xi, t)e^{-t-y}$ under the respective measures in (5.25). Choose the cutoff equal to one

on $\{\varrho \le R\}$. Since $K_\tau(t+y) \le K_\tau(t) \le C_A W(t)$ by (5.24), tightness of the weighted measures permits $R \to \infty$ and yields

$$\lim_{\tau\to\infty} \int H(\xi,t) K_\tau(t+y) \mathfrak{G}_\tau(\mathrm{d}\xi, \mathrm{d}t) = e^{-y} \int H(\xi,t) e^{-t} \mathfrak{G}(\mathrm{d}\xi, \mathrm{d}t).$$

Taking $H = 1$, $y = 0$ in (9.24) and adding the negligible complement (5.27) proves (5.29). For general H, y the shift restricts the event to $w(r_\tau)(R - \rho_\tau(U)) > y$. Division by (5.29) yields (5.30)–(5.31) jointly with independence. □

9.5. **Reusable finite-output normal forms.**

Proof of Proposition 5.6. At $z = \lambda^{-1/2} x$ the rescaled differences in (5.33) are $\sqrt{\lambda} L_{j,s}(x) + O(\|x\|^2)$. Off the null union of their linear interfaces this yields (5.35) by (5.32). Uniform positive definiteness and $f \ge f_0$ imply (5.4) with $\varrho_s(x) = \|x\|^2$ in a sufficiently small tube, and (5.5) holds for $p > c/2$. The Jacobian is $\lambda^{-c/2}$ and

$$\int_{\mathcal{C}_s} e^{-\langle H_s x, x\rangle/2} \mathrm{d}x = \frac{(2\pi)^{c/2}}{\sqrt{\det H_s}} \mathbb{P}\{Z_s \in \mathcal{C}_s\}.$$

Now (5.7) yields (5.36). □

Proof of Proposition 5.7. Taking the maximum in (5.37) yields (5.39) by the assumed bound on (5.38) and uniformity of the remainder. With $(y, w) = (\lambda^{-1} x, \lambda^{-1/2} z)$ the limiting phase is (5.38) and the Jacobian is $\lambda^{-(c+r)/2}$. The sublevel $\|x\| + \|z\|^2 \le t$ has volume proportional to $t^{(c+r)/2}$, so (5.4)–(5.5) hold for $p > (c+r)/2$. Apply Theorem 5.1. □

9.6. **The explicit two-output result.**

Proof of Theorem 6.1. In (S) and (W) the cone in Proposition 5.6 is respectively the full normal space and a half-space through the origin. Its Gaussian mass is thus 1 and 1/2 respectively, proving (6.8)–(6.9).

Under (A) put $A_i(v) = \langle H_{i,s} v, v\rangle/2$ and $x_0 = (A_2(v) - A_1(v))/(a_s + \delta_s)$. The rank-one scaling in Proposition 5.7 yields the phase

$$\begin{aligned} \Phi_s(x,v) &= \max(a_s x + A_1(v), -\delta_s x + A_2(v)) \\ &= \tfrac{1}{2}\langle \overline{H}_s v, v\rangle + \max(a_s(x - x_0), -\delta_s(x - x_0)). \end{aligned} \tag{9.25}$$

Uniform positivity and $|x_0| \le C\|v\|^2$ imply $\Phi_s(x,v) \ge c_1(|x| + \|v\|^2)$, whence (5.39) yields (6.11). Integrating on either side of x_0 yields

$$\int_{\mathbb{R}} e^{-\Phi_s(x,v)} \mathrm{d}x = (a_s^{-1} + \delta_s^{-1}) \exp(-\langle \overline{H}_s v, v\rangle/2). \tag{9.26}$$

Gaussian integration over W_s in (5.40) now yields (6.10) with constant (6.7). □

9.7. **The Gaussian covariance reduction.**

Proof of Proposition 7.1. Write $\mathbf{Z} \overset{d}{=} A_\Gamma R V$ with independent $R \sim \chi_r$ and $V \sim \sigma_{r-1}$. Then jointly

$$(Y_j)_{j=1}^d \overset{d}{=} (R^2 g_j(V))_{j=1}^d.$$

Conditioning on V yields (7.4). If some $Q_j = 0$ then $Y_j = 0$ for that index. If all $Q_j \ne 0$ then the union of their kernels meets the sphere in a null set, so q is finite almost everywhere. Since $g_j \le \|Q_j\|$ and q is lower semicontinuous on the compact sphere, $0 < q_* < \infty$. The chi tail satisfies (2.9) with $w(x) = x$, yielding (7.5). Replacing A_Γ by another full-rank factor $A_\Gamma O$, O orthogonal, conjugates each Q_j by O and preserves σ_{r-1}. □

Proof of Corollary 7.2. Write the grid vectors jointly as $B\mathbf{N}_\ell$ in law with independent $\mathbf{N}_\ell \sim N_r(0, I_r)$, $1 \le \ell \le \nu$. Stacking these vectors of standard normals yields the forms in (7.6), all nonzero since $\|b_j\|^2 = K_{jj} > 0$. By Proposition 7.1 we have $g_* = q_*^{-2} > 0$ and

$$R \sim \chi_{\nu r}, \qquad r_\tau = \sqrt{u/g_*}, \qquad \lambda_\tau = u/g_*, \qquad \tau = \sqrt{u}.$$

Thus Theorem 5.1 and the chi-tail equivalent yield (7.8)–(7.9). For degree two, (5.10) reads $\zeta_\tau = 1/(2g_*)$. The minimum, vector and location limits now follow from Corollary 5.2 under their respective hypotheses. Factor invariance follows from Proposition 7.1. □

Proof of Corollary 7.4. Gaussian regression yields

$$X_\ell(t_j) = \rho_j X_\ell(t_1) + \varepsilon_{\ell j}, \qquad 1 \le \ell \le \nu, \quad 2 \le j \le d, \tag{9.27}$$

where the residual rows are independent $N_{d-1}(0, K^\circ)$ vectors independent of $\mathbf{X}_1 = (X_\ell(t_1))_{\ell=1}^\nu$. In standard Gaussian coordinates for $\mathbf{X}_1/\sigma$ and the residuals, write an angle as $(\sqrt{1-\|z\|^2}\theta, z)$ with $\theta \in \mathbb{S}^{\nu-1}$ and $z \in \mathbb{R}^c$, $c = \nu(d-1)$. Then $g_1 = \sigma^2(1-\|z\|^2)$ and at $z = 0$ each $g_j = \rho_j^2\sigma^2 \ge \sigma^2$ for $j \ge 2$. Hence $g_* = \sigma^2$, $\mathcal{M} = \{(\theta, 0) : \theta \in \mathbb{S}^{\nu-1}\}$ and

$$f_1 = (1-\|z\|^2)^{-1/2} = 1 + \tfrac{1}{2}\|z\|^2 + o(\|z\|^2). \tag{9.28}$$

Write $\mathcal{L}_j z$ for the residual at t_j in these coordinates. Expansion of $f_j = \sigma/\|\rho_j\sigma\sqrt{1-\|z\|^2}\theta + \mathcal{L}_j z\|$ at $z = 0$ yields uniformly in θ

$$f_j - f_1 = -\frac{\rho_j}{\sigma}\langle\theta, \mathcal{L}_j z\rangle + O(\|z\|^2), \qquad j \in \mathcal{W}. \tag{9.29}$$

For standard normal z the projections $(\langle\theta, \mathcal{L}_j z\rangle)_{j=2}^d$ have covariance $K^\circ > 0$. Thus each linear form in (9.29) is nonzero and the cone probability is π_0. The remaining constraints are slack and (9.28) supplies coercivity. By Proposition 5.6

$$C_\Phi = \pi_0(2\pi)^{c/2}\frac{|\mathbb{S}^{\nu-1}|}{|\mathbb{S}^{\nu d-1}|} = \pi_0 2^{c/2}\frac{\Gamma(\nu d/2)}{\Gamma(\nu/2)}. \tag{9.30}$$

Now (7.8)–(7.9) with $\kappa = c/2$ yield (7.15)–(7.16).

Condition on $Y_1 > u$ and put $E_u = Y_1 - u$, $\Theta = \mathbf{X}_1/\|\mathbf{X}_1\|$ and $Z^\circ_{u,j} = \sum_{\ell=1}^\nu \Theta_\ell\varepsilon_{\ell j}$. Given $\mathbf{X}_1$, $(Z^\circ_{u,j})_{j=2}^d \sim N(0, K^\circ)$ and hence is independent of E_u. The chi-square tail yields $E_u \xrightarrow[u\to\infty]{d} E_0$. By (9.27)

$$Y_j - u = (\rho_j^2 - 1)u + \rho_j^2 E_u + 2\rho_j\sqrt{u + E_u}Z^\circ_{u,j} + \sum_{\ell=1}^\nu \varepsilon_{\ell j}^2. \tag{9.31}$$

The last sum is tight under this conditioning. Dividing by $2\sqrt{u}$ on $\mathcal{W}$ and by u on $\mathcal{S}_0$ yields the joint limit in (7.17) with $\mathbf{Z}^\circ$ in place of $\mathbf{Z}^{\circ,+}$. The slack limits are positive and the weak inequalities have a null boundary and probability $\pi_0 > 0$. Restricting this convergence to the positive orthant and dividing by its probability proves (7.17). All later excesses thus tend to infinity in conditional probability while $Y_1 - u$ is tight, which proves (7.18). □

9.8. **The finite-time Brownian result.**

Proof of Corollary 7.9. The distinct positive times yield a positive-definite Brownian covariance matrix by independence of the successive increments. With $K_{ij} = \min(t_i, t_j)$ we have $\sigma^2 = t_1$, $\rho_j = 1$ for $j \ge 2$ and $K^\circ_{ij} = \min(t_i - t_1, t_j - t_1)$ for $i, j \ge 2$. Hence $\mathbf{Z}^\circ \overset{d}{=} (W(t_j - t_1))_{j=2}^d$ and $\pi_0 = \pi_d(\mathbf{T})$. The assertions follow from (7.15)–(7.18). □

Appendix A. Verification criteria for weighted angular laws

We use the tubular coordinates of Section 5.1.

Lemma A.1. *Suppose that μ has locally integrable density ω in the tube*

$$\mu(\mathrm{d}\Psi(s,z)) = \omega(s,z)\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}z. \tag{A.1}$$

Put

$$g_\tau(s,x) = \frac{|\det D_{\lambda_\tau,s}|}{a_\tau}\omega\big(s, D_{\lambda_\tau,s}x\big)\mathbf{1}\{\Psi(s, D_{\lambda_\tau,s}x) \in \mathcal{T}\} \tag{A.2}$$

and set

$$\widetilde{\phi}_\tau(s,x) := \phi_\tau\big(\Psi(s, D_{\lambda_\tau,s}x)\big)$$

on the rescaled tube, with arbitrary values elsewhere. Assume $g_\tau \to g_0$ in $L^1_{\mathrm{loc}}(\mathsf{N}\mathcal{M})$ as $\tau\to\infty$ and

$$\widetilde{\phi}_\tau(s,x) \xrightarrow{\tau\to\infty} \Phi(s,x) \tag{A.3}$$

locally in $g_\tau \mathrm{d}V_{\mathcal{M}}\mathrm{d}x$-measure for a Borel map $\Phi : \mathsf{N}\mathcal{M} \to [-A,\infty]$. Thus for each compact K and each $\varepsilon > 0$

$$\int_K g_\tau(s,x)\mathbf{1}\left\{d_\infty\Big(\widetilde{\phi}_\tau(s,x), \Phi(s,x)\Big) > \varepsilon\right\}\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x \xrightarrow{\tau\to\infty} 0, \tag{A.4}$$

with $d_\infty(s,t) = |e^{-s} - e^{-t}|$. Then (5.25) holds with

$$\mathfrak{G}(\mathrm{d}s,\mathrm{d}x,\mathrm{d}t) = g_0(s,x)\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x\delta_{\Phi(s,x)}(\mathrm{d}t). \tag{A.5}$$

If for some $p > 0$ a continuous coercive gauge ϱ satisfies

$$\lim_{R\to\infty}\limsup_{\tau\to\infty}\int_{\{\varrho(s,x)>R\}}\big(1 + (\widetilde{\phi}_\tau(s,x))_+\big)^{-p} g_\tau(s,x)\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x = 0, \tag{A.6}$$

then (5.26) holds and

$$C_{\mathfrak{G}} = \int_{\mathsf{N}\mathcal{M}} g_0(s,x)e^{-\Phi(s,x)}\mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x. \tag{A.7}$$

Lemma A.2. *Let B be an $r\times k_0$ matrix of full row rank and let $\gamma_1,\dots,\gamma_r \in (-1,0)$. For every $M < \infty$*

$$\sup_{\mathbf{a}\in\mathbb{R}^r}\int_{[-M,M]^{k_0}}\prod_{i=1}^r |(Bx+\mathbf{a})_i|^{\gamma_i}\mathrm{d}x < \infty. \tag{A.8}$$

The bound is uniform for B in compact subsets of the full-row-rank locus.

Proposition A.3. *Let*

$$\omega(s,z) = \omega_0(s,z)\prod_{\ell=1}^N |\psi_\ell(s,z)|^{\beta_\ell}, \qquad \beta_\ell > -1, \tag{A.9}$$

in the tube, with ω_0 continuous and positive near the zero section. Suppose $|\det D_{\lambda,s}| = d_\lambda$ is independent of s and for some $c_{\ell,\lambda} > 0$

$$c_{\ell,\lambda}^{-1}\psi_\ell(s, D_{\lambda,s}x) \xrightarrow{\lambda\to\infty} L_{\ell,s}(x) \tag{A.10}$$

locally uniformly off the zero sets of the limiting forms, whose union has $\mathrm{d}V_{\mathcal{M}}\mathrm{d}x$-measure zero. Put

$$a_\lambda = d_\lambda\prod_{\ell=1}^N c_{\ell,\lambda}^{\beta_\ell} \tag{A.11}$$

and

$$H_\lambda(s,x) = \mathbf{1}\{\Psi(s, D_{\lambda,s}x) \in \mathcal{T}\} \prod_{\ell=1}^{N} \left| c_{\ell,\lambda}^{-1} \psi_\ell(s, D_{\lambda,s}x) \right|^{\beta_\ell}.$$

If for some λ_0 *and every compact* $K \subset \mathsf{N}\mathcal{M}$

$$\lim_{M\to\infty} \sup_{\lambda\geq\lambda_0} \int_K H_\lambda(s,x)\mathbf{1}\{H_\lambda(s,x) > M\} \mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x = 0 \tag{A.12}$$

then the densities in (A.2) *with* $a_\tau = a_{\lambda_\tau}$ *converge in* L^1_{loc} *to*

$$g_0(s,x) = \omega_0(s,0) \prod_{\ell=1}^{N} |L_{\ell,s}(x)|^{\beta_\ell}. \tag{A.13}$$

To verify (A.12) suppose the normalised negative-power factors before passage to the limit are uniformly comparable in absolute value to the components of $B_{\lambda,s}x + \mathbf{a}_{\lambda,s}$. If $B_{\lambda,s}$ ranges over a compact subset of the full-row-rank locus and the nonnegative-power factors are locally uniformly bounded, Lemma A.2 bounds the local L^q norms of H_λ for $q > 1$ sufficiently close to one. In the triangular case the same argument applies block by block provided the remaining singular factors are independent of the block just integrated.

Separate rank conditions on overlapping blocks allow the nonintegrable product $|x|^{-3/4}|x|^{-3/4}$. Full rank of the limiting forms alone also allows concentration near their zero sets.

For (A.6) suppose $\widetilde{\phi}_\tau \geq c\varrho - C$ with $c > 0$, each unit cube $\mathcal{Q}$ has rescaled angular mass at most $C(1 + \inf_{\mathcal{Q}} \varrho)^B$ and at most $C(1+R)^D$ cubes meet $\{\varrho \leq R\}$. Summation over dyadic annuli yields (A.6) for every $p > B + D$ in (5.24).

A.1. **Proofs of the verification criteria.**

Proof of Lemma A.1. Write $\mathrm{d}m(\xi) = \mathrm{d}V_{\mathcal{M}}(s)\mathrm{d}x$ for $\xi = (s,x)$. For $f \in C_c(\mathsf{N}\mathcal{M} \times \mathbb{I}_A)$, (A.2) yields

$$\int f \mathrm{d}\mathfrak{G}_\tau = \int f(\xi, \widetilde{\phi}_\tau(\xi)) g_\tau(\xi) \mathrm{d}m(\xi) \longrightarrow \int f(\xi, \Phi(\xi)) g_0(\xi) \mathrm{d}m(\xi), \qquad \tau \to \infty.$$

Indeed f is uniformly continuous on the compact projection of its support times $\mathbb{I}_A$. By (A.4) and local L^1 boundedness the phase replacement costs $o(1)$, and $g_\tau \to g_0$ in L^1_{loc}. This proves (A.5), whence (5.28) yields (A.7). Pulling back the integral in (5.26) by the same change of variables yields exactly (A.6). □

Proof of Lemma A.2. Complete B_0 of full row rank to an invertible matrix $T_{B_0} = \binom{B_0}{C}$. For B near B_0, $|\det T_B|^{-1}$ is bounded and $T_B[-M,M]^{k_0} \subset [-M',M']^{k_0}$ with $T_B = \binom{B}{C}$. Changing to $z = T_B x$ bounds the integral in (A.8) by a constant times

$$\prod_{i=1}^{r} \sup_{a\in\mathbb{R}} \int_{-M'}^{M'} |z+a|^{\gamma_i} \mathrm{d}z = \prod_{i=1}^{r} \frac{2(M')^{1+\gamma_i}}{1+\gamma_i},$$

since $-1 < \gamma_i < 0$ and the centred interval maximises each integral. A finite cover yields uniformity on compact sets of full-row-rank matrices. □

Proof of Proposition A.3. By (A.2), (A.9) and (A.11), $g_\lambda(s,x) = \omega_0(s, D_{\lambda,s}x) H_\lambda(s,x)$. On every compact set the tube indicator is eventually one by uniform contraction, and the continuous factor tends uniformly to $\omega_0(s,0)$. Thus (A.10) yields convergence to (A.13) off the null union of limiting zero sets. The uniform integrability in (A.12) passes to g_λ by local boundedness of the continuous factor, so Vitali's theorem yields convergence in L^1_{loc}. □

Appendix B. Dirichlet angular specialisation

Let U have the symmetrised Dirichlet law on $\mathbb{S}^{k-1}$, $k \geq 2$, with parameter $\boldsymbol{\alpha} \in (0,\infty)^k$ and surface density [5]

$$h_{\boldsymbol{\alpha}}(u) = c_{\boldsymbol{\alpha}} \prod_{i=1}^{k} |u_i|^{\beta_i}, \qquad \beta_i = 2\alpha_i - 1, \qquad c_{\boldsymbol{\alpha}} = \frac{\Gamma(\sum_{i=1}^k \alpha_i)}{2\prod_{i=1}^k \Gamma(\alpha_i)}. \tag{B.1}$$

Consider

$$X = A^\top R U, \qquad C = (A^\top)^{-1}, \qquad \Sigma = A^\top A, \tag{B.2}$$

where A is nonsingular and the radius R with law F is independent of U. For $\mathbf{b} \in (0,\infty)^k$ let $\widetilde{\mathbf{b}}$ minimise the strictly convex program

$$\min_{x \geq \mathbf{b}} x^\top \Sigma^{-1} x. \tag{B.3}$$

The KKT conditions for $\boldsymbol{\eta} = \Sigma^{-1}\widetilde{\mathbf{b}}$ are $\boldsymbol{\eta} \geq \mathbf{0}$ and $\eta_i(\widetilde{b}_i - b_i) = 0$ for $1 \leq i \leq k$. Thus $I = \{i : \eta_i > 0\}$ with complement J is the unique minimal index set satisfying [6]

$$\widetilde{\mathbf{b}}_I = \mathbf{b}_I, \qquad \widetilde{\mathbf{b}}_J = \Sigma_{JI}\Sigma_{II}^{-1}\mathbf{b}_I \geq \mathbf{b}_J, \qquad \Sigma_{II}^{-1}\mathbf{b}_I > \mathbf{0}_I. \tag{B.4}$$

Put $m_I = |I|$ and

$$\mu_* = \left(\mathbf{b}_I^\top \Sigma_{II}^{-1}\mathbf{b}_I\right)^{1/2}, \qquad u_* = \frac{C\widetilde{\mathbf{b}}}{\mu_*} \in \mathbb{S}^{k-1}. \tag{B.5}$$

For $X > u\mathbf{b}$ the minimum cost $q_* = \mu_*$ is attained only at u_*.

Suppose F satisfies (2.9) with auxiliary function w. For $u_n \to \infty$ put $r_n = \mu_* u_n$, $w_n = w(r_n)$ and $\lambda_n = r_n w_n$. For any $A_0, p > 0$ the Gumbel bound yields $C_{A_0,p} < \infty$ such that

$$\frac{\overline{F}(r_n + t/w_n)}{\overline{F}(r_n)} \leq C_{A_0,p}(1 + t_+)^{-p}, \qquad t \geq -A_0, \tag{B.6}$$

for all sufficiently large n. Moving thresholds on the natural scales are assumed eventually componentwise positive and satisfy

$$w_n(\mathbf{t}_n - u_n\widetilde{\mathbf{b}})_I \xrightarrow{n\to\infty} \boldsymbol{\xi}_I \in \mathbb{R}^{m_I}, \tag{B.7}$$

$$\left(\frac{w_n}{r_n}\right)^{1/2} (\mathbf{t}_n - u_n\widetilde{\mathbf{b}})_J \xrightarrow{n\to\infty} \boldsymbol{\xi}_J \in [-\infty,\infty)^{|J|}. \tag{B.8}$$

Only residual coordinates with $\xi_j = -\infty$ require a separate positivity assumption. Put

$$L = \{i : \beta_i \neq 0,\ (C\widetilde{\mathbf{b}})_i = 0\}, \tag{B.9}$$

$$L_J^0 = \{i \in L : J \neq \emptyset,\ C_{iJ} \neq \mathbf{0}_J\}, \qquad L_I^0 = L \setminus L_J^0. \tag{B.10}$$

Put

$$L_J^- = \{i \in L_J^0 : \beta_i < 0\}, \qquad L_I^- = \{i \in L_I^0 : \beta_i < 0\}.$$

We shall assume that $C_{L_J^-,J}$ has full row rank, with no restriction if $L_J^- = \emptyset$. The rows indexed by L_I^- vanish on J, hence $C_{L_I^-,I}$ has full row rank by nonsingularity of C. The chart has $m_I - 1$ linear and $|J|$ quadratic directions, whence

$$d_\lambda = \lambda^{-(m_I-1)-|J|/2}. \tag{B.11}$$

A zero Dirichlet factor indexed by L_J^0 first appears on the $\lambda^{-1/2}$ scale, whereas one indexed by L_I^0 first appears on the λ^{-1} scale. Proposition A.3 therefore yields

$$\begin{aligned} a_\lambda &= \lambda^{-(m_I-1)-|J|/2-\frac{1}{2}\sum_{i\in L_J^0}\beta_i - \sum_{i\in L_I^0}\beta_i} \\ &= \lambda^{\kappa_D}, \end{aligned} \tag{B.12}$$

$$\kappa_D = 1 - m_I - \frac{|J|}{2} + \frac{|L_J^0|}{2} + |L_I^0| - \sum_{i\in L_J^0} \alpha_i - 2\sum_{i\in L_I^0} \alpha_i. \tag{B.13}$$

For $L = \emptyset$ we recover $\kappa_D = -(m_I - 1) - |J|/2$. Define

$$\tau^*_{\mathrm{reg}} = \prod_{\substack{i\notin L\\ \beta_i\neq 0}} \mu_*^{-\beta_i} |(C\widetilde{\mathbf{b}})_i|^{\beta_i}, \tag{B.14}$$

$$H(\mathbf{y}) = \prod_{i\in L_I^0} |C_{iI}\mathbf{y}_I|^{\beta_i} \prod_{i\in L_J^0} |C_{iJ}\mathbf{y}_J|^{\beta_i}, \tag{B.15}$$

$$S(\mathbf{y}) = \frac{\mathbf{b}_I^\top}{\mu_*}\Sigma_{II}^{-1}\mathbf{y}_I + \frac{1}{2}\mathbf{y}_J^\top(\Sigma^{-1})_{JJ}\mathbf{y}_J. \tag{B.16}$$

For $\mathbf{z}_I \in \mathbb{R}^{m_I}$ and $\mathbf{z}_J \in [-\infty,\infty)^{|J|}$ put

$$K_D(\mathbf{z}) = \int_{\mathbf{y}>\mathbf{z}} H(\mathbf{y}) e^{-S(\mathbf{y})} \mathrm{d}\mathbf{y}.$$

A lower bound $-\infty$ imposes no restriction.

Lemma B.1. *Let*

$$\widehat{\mathbf{b}}_I = \frac{\mathbf{b}_I}{\mu_*}, \qquad \boldsymbol{\ell} = \Sigma_{II}^{-1}\widehat{\mathbf{b}}_I, \qquad Q = (\Sigma^{-1})_{JJ}, \qquad \mathcal{E} = \{v_I : \boldsymbol{\ell}^\top v_I = 0\}. \tag{B.17}$$

Use the chart

$$U_\lambda(v_I, v_J) = \frac{u_* + C\binom{\lambda^{-1}v_I}{\lambda^{-1/2}v_J}}{\|u_* + C\binom{\lambda^{-1}v_I}{\lambda^{-1/2}v_J}\|}, \qquad v_I \in \mathcal{E}. \tag{B.18}$$

Then $\boldsymbol{\ell}^\top\widehat{\mathbf{b}}_I = 1$. *For* $v_J > \boldsymbol{\xi}_J$ *put*

$$\tau_{\boldsymbol{\xi}}(v_I) = \max_{i\in I} \frac{\xi_i - v_i}{\widehat{b}_i}, \qquad \Phi_{\boldsymbol{\xi}}(v_I, v_J) = \tau_{\boldsymbol{\xi}}(v_I) + \frac{1}{2}v_J^\top Q v_J. \tag{B.19}$$

If $R = r_n + s/w_n$, *then, as* $n\to\infty$, *locally uniformly in the chart*

$$w_n(A^\top R U_{\lambda_n} - u_n\widetilde{\mathbf{b}})_I = v_I + \widehat{\mathbf{b}}_I\left(s - \frac{1}{2}v_J^\top Q v_J\right) + o(1), \tag{B.20}$$

$$\left(\frac{w_n}{r_n}\right)^{1/2}(A^\top R U_{\lambda_n} - u_n\widetilde{\mathbf{b}})_J = v_J + o(1). \tag{B.21}$$

The limiting radial phase is $\Phi_{\boldsymbol{\xi}}$ *on* $v_J > \boldsymbol{\xi}_J$ *and infinite if some* $v_j < \xi_j$. *On the null interfaces* $v_j = \xi_j$ *with finite* ξ_j *we assign it the value infinity. Put*

$$\mathcal{H}(v_I, v_J) = \prod_{i\in L_I^0} |C_{iI}v_I|^{\beta_i} \prod_{i\in L_J^0} |C_{iJ}v_J|^{\beta_i}. \tag{B.22}$$

With Euclidean surface measure $\mathrm{d}\sigma_{\mathcal{E}}$ *we have*

$$\begin{aligned} &\frac{1}{|\Sigma|^{1/2}\|\boldsymbol{\ell}\|}\int_{v_J>\boldsymbol{\xi}_J}\int_{\mathcal{E}} \mathcal{H}(v_I, v_J) e^{-\Phi_{\boldsymbol{\xi}}(v_I,v_J)}\mathrm{d}\sigma_{\mathcal{E}}(v_I)\mathrm{d}v_J \\ &\qquad = \frac{K_D(\boldsymbol{\xi})}{|\Sigma|^{1/2}}. \end{aligned} \tag{B.23}$$

Lemma B.2. *Let* ρ_n *be the required radius for* $\mathbf{t}_n$ *and put*

$$\phi_n = w_n(\rho_n - r_n), \qquad \varrho_D(v_I, v_J) = 1 + \|v_I\| + \|v_J\|^2.$$

Let g_n *denote the density of* $a_{\lambda_n}^{-1}U_{\lambda_n}^*\mu$ *relative to* $\mathrm{d}\sigma_{\mathcal{E}}\mathrm{d}v_J$, *extended by zero outside the chart. On a sufficiently small fixed tube there are* $c, C > 0$ *and* $B \geq 0$ *such that for all large* n

$$\phi_n\big(U_{\lambda_n}(v_I, v_J)\big) \geq c\varrho_D(v_I, v_J) - C, \tag{B.24}$$

$$\int_{\mathcal{Q}} g_n(v_I, v_J)\mathrm{d}\sigma_{\mathcal{E}}(v_I)\mathrm{d}v_J \le C\left(1+\inf_{\mathcal{Q}} \varrho_D\right)^B \tag{B.25}$$

for every unit cube $\mathcal{Q} \subset \mathcal{E} \times \mathbb{R}^J$.

Corollary B.3. *For the Dirichlet model above suppose* $C_{L_J^-,J}$ *has full row rank and the eventually positive thresholds satisfy* (B.7)–(B.8). *Then*

$$\mathbb{P}\{X > \mathbf{t}_n\} \underset{n\to\infty}{\sim} \frac{c_{\boldsymbol{\alpha}}\tau^*_{\mathrm{reg}}}{|\Sigma|^{1/2}}\lambda_n^{\kappa_D}\overline{F}(r_n)K_D(\boldsymbol{\xi}). \tag{B.26}$$

Define the componentwise scales

$$(\mathbf{d}_n)_I = w_n\mathbf{1}_I, \qquad (\mathbf{d}_n)_J = \left(\frac{w_n}{r_n}\right)^{1/2}\mathbf{1}_J. \tag{B.27}$$

At each continuity point $\mathbf{x} \in [0,\infty)^k$ *we have*

$$\mathbb{P}\{\mathbf{d}_n(X-\mathbf{t}_n) > \mathbf{x} | X > \mathbf{t}_n\} \xrightarrow{n\to\infty} \frac{K_D(\boldsymbol{\xi}+\mathbf{x})}{K_D(\boldsymbol{\xi})}. \tag{B.28}$$

The limiting I- *and* J-*blocks are independent. If* $\xi_j = -\infty$ *the scaled residual excess diverges in conditional probability and we take* $-\infty + x_j = -\infty$. *If* $L_I^0 = \emptyset$ *then*

$$\frac{K_D(\boldsymbol{\xi}+\mathbf{x})}{K_D(\boldsymbol{\xi})} = \exp(-\boldsymbol{\ell}^\top\mathbf{x}_I)\frac{K_D(\boldsymbol{\xi}_I, \boldsymbol{\xi}_J+\mathbf{x}_J)}{K_D(\boldsymbol{\xi})}. \tag{B.29}$$

The active excesses then have independent exponential limits with rates ℓ_i, $i \in I$. *The residual factor is one when* $J = \emptyset$.

Remark B.4. *With* $\gamma_j = 1$, $G_j(x) = (A^\top x)_j$ *and* $\boldsymbol{\xi} = \mathbf{q}$ *in [7, Theorem 3.1 and Corollary 4.1],* (B.26)–(B.29) *recover the Dirichlet tail equivalent and conditional excess law with the same constant and rank assumptions.*

If the negative-power forms lose rank, (5.23) may fail to be tight under (B.12). Coalescing factors can yield a normalisation such as $\lambda^{-1}\log\lambda$. To apply Theorem 5.5 one must verify graph convergence, tightness, the complement estimate and positivity of the limiting Laplace integral for that normalisation.

B.1. **Proofs of the Dirichlet specialisation.**

Proof of Lemma B.1. By (B.4)–(B.5) and (B.17), $\boldsymbol{\ell}^\top\widehat{\mathbf{b}}_I = 1$ and

$$\Sigma^{-1}\widetilde{\mathbf{b}} = \mu_*(\boldsymbol{\ell}, \mathbf{0}_J)^\top, \qquad C^\top u_* = (\boldsymbol{\ell}, \mathbf{0}_J)^\top. \tag{B.30}$$

For $v_I \in \mathcal{E}$ the denominator D_n of (B.18) with $\lambda = \lambda_n$ therefore satisfies locally uniformly

$$D_n = 1 + \frac{v_J^\top Q v_J}{2\lambda_n} + O(\lambda_n^{-3/2}).$$

Substitution into (B.18) with $A^\top C = I$ yields (B.20)–(B.21). By (B.7) the maximum active phase tends to $\tau_{\boldsymbol{\xi}}(v_I) + v_J^\top Q v_J/2$.

For $j \in J$ put $\widehat{b}_j = \widetilde{b}_j/\mu_*$, $\xi_{n,j} = (w_n/r_n)^{1/2}(t_{n,j} - u_n\widetilde{b}_j)$ and $\phi_{n,j} = w_n(\rho_{n,j} - r_n)$ for the required branch radius $\rho_{n,j}$. On compact sets the branch denominator is eventually positive and

$$\phi_{n,j}(U_{\lambda_n}(v_I, v_J)) = \lambda_n(D_n - 1) + D_n\frac{\sqrt{\lambda_n}(\xi_{n,j} - v_j)}{\widehat{b}_j + \lambda_n^{-1/2}v_j}. \tag{B.31}$$

By (B.8) this tends locally uniformly to $-\infty$ for $v_j > \xi_j$ and to $+\infty$ for $v_j < \xi_j$, including the first case throughout if $\xi_j = -\infty$. Finite interfaces are null.

Insert

$$e^{-\Phi} = \int_0^\infty e^{-(\Phi+e)}\mathrm{d}e \tag{B.32}$$

and map $v_I \in \mathcal{E}$, $v_J > \boldsymbol{\xi}_J$, $e > 0$ to the orthant $\mathbf{y} > \boldsymbol{\xi}$ by

$$\mathbf{y}_J = v_J, \qquad \mathbf{y}_I = v_I + \widehat{\mathbf{b}}_I(\tau_{\boldsymbol{\xi}}(v_I) + e). \tag{B.33}$$

The inverse is

$$c = \boldsymbol{\ell}^\top \mathbf{y}_I, \qquad v_I = \mathbf{y}_I - \widehat{\mathbf{b}}_I c, \qquad e = \min_{i\in I} \frac{y_i - \xi_i}{\widehat{b}_i}, \qquad v_J = \mathbf{y}_J. \tag{B.34}$$

Translation from e to c at fixed v_I and $\boldsymbol{\ell}^\top \widehat{\mathbf{b}}_I = 1$ yield

$$\mathrm{d}\sigma_{\mathcal{E}}(v_I)\mathrm{d}e = \|\boldsymbol{\ell}\|\mathrm{d}\mathbf{y}_I, \qquad \Phi_{\boldsymbol{\xi}}(v_I, v_J) + e = S(\mathbf{y}). \tag{B.35}$$

Since $C_{iI}\widehat{\mathbf{b}}_I = 0$ for $i \in L_I^0$, we have $\mathcal{H}(v_I, v_J) = H(\mathbf{y})$, proving (B.23). For the unit normal $n = (\boldsymbol{\ell}, \mathbf{0}_J)^\top/\|\boldsymbol{\ell}\|$ the tangential Jacobian of C is $|\det C|\|C^{-\top}n\| = |\Sigma|^{-1/2}/\|\boldsymbol{\ell}\|$ by (B.30). □

Proof of Lemma B.2. Put $z_n = \binom{\lambda_n^{-1} v_I}{\lambda_n^{-1/2} v_J}$ and $D_n = \|u_* + Cz_n\|$. By (B.30) and $v_I \in \mathcal{E}$

$$D_n^2 = 1 + z_n^\top \Sigma^{-1} z_n. \tag{B.36}$$

For $i \in I$ let $\rho_{n,i}$ be the radius required by the ith threshold and put $\xi_{n,i} = w_n(t_{n,i} - u_n b_i)$. Since $A^\top C = I$ and $A^\top u_* = \widetilde{\mathbf{b}}/\mu_*$

$$\frac{\rho_{n,i}(U_{\lambda_n})}{r_n} = D_n \frac{\lambda_n \widehat{b}_i + \xi_{n,i}}{\lambda_n \widehat{b}_i + v_i}. \tag{B.37}$$

On $\mathcal{E}$ the maximum of $-v_i/\widehat{b}_i$ over $i \in I$ is positive off zero since $\boldsymbol{\ell} > 0$ and $\boldsymbol{\ell}^\top v_I = 0$. By homogeneity, compactness of the unit sphere and boundedness in (B.7)

$$\max_{i\in I} \frac{\xi_{n,i} - v_i}{\widehat{b}_i} \geq c_1\|v_I\| - C_1, \tag{B.38}$$

also when $\mathcal{E} = \{0\}$. In a sufficiently small fixed tube $1 \leq D_n \leq 2$ and $1 + v_i/(\lambda_n \widehat{b}_i) \in [1/2, 2]$. Thus (B.36) yields $\lambda_n(D_n - 1) \geq c_2\|v_J\|^2$, and (B.37)–(B.38) yield

$$\phi_n(U_{\lambda_n}) \geq \lambda_n(D_n - 1) + D_n \max_{i\in I} \frac{\lambda_n(\xi_{n,i} - v_i)}{\lambda_n \widehat{b}_i + v_i} \geq c_3(\|v_I\| + \|v_J\|^2) - C_3.$$

This proves (B.24).

The tangential Jacobian in the proof of Lemma B.1, the dilation (B.11) and central projection yield the surface Jacobian $d_{\lambda_n} D_n^{-k}/(|\Sigma|^{1/2}\|\boldsymbol{\ell}\|)$. With $2\bar{\alpha} = k + \sum_{i=1}^k \beta_i$, (B.1) and (B.12) yield

$$\begin{aligned} g_n(v_I, v_J) = {} & \frac{c_{\boldsymbol{\alpha}} D_n^{-2\bar{\alpha}}}{|\Sigma|^{1/2}\|\boldsymbol{\ell}\|} \prod_{\substack{i\notin L \\ \beta_i \neq 0}} |u_{*,i} + (Cz_n)_i|^{\beta_i} \\ & \times \prod_{i\in L_I^0} |C_{iI} v_I|^{\beta_i} \prod_{i\in L_J^0} |C_{iJ} v_J + \lambda_n^{-1/2} C_{iI} v_I|^{\beta_i} \end{aligned} \tag{B.39}$$

on the chart and $g_n = 0$ elsewhere. Shrink the tube so that the regular coordinates stay bounded away from zero. On a unit cube $\mathcal{Q}$ the nonnegative powers in the last two products are bounded together by $C(1+\inf_{\mathcal{Q}} \varrho_D)^B$ with $B = \sum_{i\in L} \max(\beta_i, 0)$. For the negative powers apply Lemma A.2 first in v_J using full row rank of $C_{L_J^-, J}$ and uniformity in the shift $\lambda_n^{-1/2} C_{iI} v_I$. Then apply it in $v_I \in \mathcal{E}$. Indeed $C_{L_I^-, I}$ has independent rows annihilating $\widehat{\mathbf{b}}_I$ and $\mathbb{R}^I = \mathcal{E} \oplus \operatorname{span}(\widehat{\mathbf{b}}_I)$, so their restrictions to $\mathcal{E}$ remain independent. This proves (B.25). □

Proof of Corollary B.3. Apply Theorem 5.5 to $G_j(x) = (A^\top x)_j$, $1 \le j \le k$. Choose $q > 1$ with $q\beta_i > -1$ for every negative β_i, $i \in L$. On each compact $K \subset \mathcal{E} \times \mathbb{R}^J$ the regular and nonnegative-power factors in (B.39) are bounded uniformly. For the negative powers apply Lemma A.2 with exponents $q\beta_i$ first in v_J, uniformly in the shift depending on v_I, and then in $v_I \in \mathcal{E}$. Both ranks are full by the hypothesis on $C_{L_J^-,J}$ and the restriction argument in the proof of Lemma B.2. Thus $\sup_{\lambda\ge\lambda_0} \int_K H_\lambda^q \mathrm{d}\sigma_{\mathcal{E}}\mathrm{d}v_J < \infty$, which implies (A.12). By Proposition A.3, $g_n \to g_0$ in L^1_{loc}. The phase converges off the finite interface hyperplanes by Lemma B.1. These are Lebesgue-null, so on each compact the convergence also holds in $g_n \mathrm{d}\sigma_{\mathcal{E}}\mathrm{d}v_J$-measure by local L^1 convergence. The graph limit follows from Lemma A.1.

Put $D = m_I - 1 + |J|/2$ and choose $p > \max(B + D + 1, -\kappa_D)$ in (B.6). By (B.24) the phases are bounded below on the tube and $(1 + (\phi_n)_+)^{-p} \le C\varrho_D^{-p}$. At most $O(R^D)$ unit cubes of a fixed lattice meet $\{\varrho_D \le R\}$ and $\sup_{\mathcal{Q}} \varrho_D \le C \inf_{\mathcal{Q}} \varrho_D$ on each cube. Summing (B.25) over dyadic shells yields (5.26). The graph integral is finite by this bound and equals

$$C_{\mathfrak{G}} = \frac{c_{\boldsymbol{\alpha}} \tau^*_{\mathrm{reg}}}{|\Sigma|^{1/2}} K_D(\boldsymbol{\xi})$$

by (A.7), (B.23) and (B.39). It is positive since $H > 0$ almost everywhere in the nonempty orthant.

Let q_I be the fixed angular cost in (2.3) restricted to $i \in I$. By (B.4) the active constraints alone have the same unique minimiser in (B.3), so q_I has minimum μ_* only at u_*. Lower semicontinuity and compactness yield $\inf_{\theta\notin\mathcal{T}} q_I(\theta) \ge (1 + 2\eta)\mu_*$ for some $\eta > 0$. Since $t_{n,i}/(u_n b_i) \to 1$ uniformly over $i \in I$ by (B.7), we have $\rho_n \ge (1 + \eta)r_n$ off $\mathcal{T}$ for all large n. Thus (B.6) bounds the complement by $O(\lambda_n^{-p}\overline{F}(r_n)) = o(\lambda_n^{\kappa_D}\overline{F}(r_n))$. This proves (5.27) and hence (B.26).

Conditioning expresses the left-hand side of (B.28) as the tail at $\mathbf{t}_n + \mathbf{x}/\mathbf{d}_n$ divided by the tail at $\mathbf{t}_n$ with componentwise division. By (B.7), (B.8) and (B.27) the shifted limits are $\boldsymbol{\xi} + \mathbf{x}$. Taking the ratio in (B.26) proves (B.28), including divergence of each residual excess with $\xi_j = -\infty$ since a finite shift leaves that lower bound unchanged. Both the orthant and the integrand in K_D factor over I and J, proving block independence. If $L_I^0 = \emptyset$ the active integral factors over $i \in I$ and its ratio is $\exp(-\boldsymbol{\ell}^\top \mathbf{x}_I)$, yielding (B.29). □

Appendix C. A two-time squared Brownian norm

For independent standard Brownian motions B_1, B_2 write $B(r) = (B_1(r), B_2(r))$ and put

$$Y(r) = B_1^2(r) + B_2^2(r), \qquad r > 0. \tag{C.1}$$

For $0 < s < t$ put $\Delta = t - s$ and

$$p_b(u) = \mathbb{P}\{Y(s) > u,\ Y(t) > bu\}, \qquad b > 0. \tag{C.2}$$

C.1. **Radial representation.** For a standard Gaussian vector $Z = (Z_1, Z_2, Z_3, Z_4)$ we may realise

$$B(s) = \sqrt{s}(Z_1, Z_2), \tag{C.3}$$

$$B(t) = \sqrt{s}(Z_1, Z_2) + \sqrt{\Delta}(Z_3, Z_4). \tag{C.4}$$

Thus $(Y(s), Y(t)) = (G_1(Z), G_2(Z))$ with

$$G_1(Z) = s(Z_1^2 + Z_2^2), \tag{C.5}$$

$$G_2(Z) = (\sqrt{s}Z_1 + \sqrt{\Delta}Z_3)^2 + (\sqrt{s}Z_2 + \sqrt{\Delta}Z_4)^2. \tag{C.6}$$

Writing $Z = RU$ yields both outputs with the same radius R, where $R^2 \sim \chi_4^2$ and U is independent of R and uniform on $\mathbb{S}^3$. At $b = 1$ the minimising set is

$$\mathcal{M} = \{(v_1, v_2, 0, 0) : v_1^2 + v_2^2 = 1\} \cong \mathbb{S}^1. \tag{C.7}$$

C.2. **Fixed-ray asymptotics.**

Theorem C.1. *For every fixed $b > 0$, as $u \to \infty$*

$$p_b(u) \sim \begin{cases} \exp\left(-\dfrac{u}{2s}\right), & 0 < b < 1, \\ \dfrac{1}{2}\exp\left(-\dfrac{u}{2s}\right), & b = 1, \\ C_b u^{-1/2}\exp(-uI_b), & 1 < b < (t/s)^2, \\ \dfrac{1}{2}\exp\left(-\dfrac{bu}{2t}\right), & b = (t/s)^2, \\ \exp\left(-\dfrac{bu}{2t}\right), & b > (t/s)^2, \end{cases} \tag{C.8}$$

where for $1 < b < (t/s)^2$

$$I_b = \frac{1}{2s} + \frac{(\sqrt{b}-1)^2}{2\Delta} \tag{C.9}$$

and

$$C_b = \frac{\Delta^{3/2} b^{1/4}}{\sqrt{2\pi}(\sqrt{b}-1)(t - s\sqrt{b})}. \tag{C.10}$$

At $b = 1$ this yields

$$\mathbb{P}\{\min(Y(s), Y(t)) > u\} \underset{u\to\infty}{\sim} \frac{1}{2}\exp\left(-\frac{u}{2s}\right). \tag{C.11}$$

Remark C.2. *For $1 < b < (t/s)^2$ put $\vartheta = \sqrt{b}$ and*

$$\alpha_b = (2sI_b)^{-1/2}, \qquad \beta_b = (\vartheta - 1)(2\Delta I_b)^{-1/2}.$$

In geodesic normal coordinates about the minimising circle $\mathcal{M}_b = \{(\alpha_b v, \beta_b v) : v \in \mathbb{S}^1\}$ we have

$$a = \frac{\sqrt{s}(\vartheta - 1)}{\sqrt{\Delta}}, \qquad \delta = \frac{t - s\vartheta}{\vartheta\sqrt{s\Delta}},$$
$$H_1 = 1 - a^2, \qquad H_2 = 1 - \delta^2, \qquad \overline{H} = 1 - a\delta = \frac{2sI_b}{\vartheta},$$
$$a^{-1} + \delta^{-1} = \frac{2\sqrt{s}\Delta^{3/2} I_b}{(\vartheta - 1)(t - s\vartheta)}.$$

Since $c = 2$, $\mathcal{M}_b$ has length 2π and the angular density is $1/(2\pi^2)$, Theorem 6.1(A) yields

$$C_{\mathrm{A}} = \frac{2}{\sqrt{\pi}} \frac{\Delta^{3/2} b^{1/4}\sqrt{I_b}}{(\sqrt{b}-1)(t - s\sqrt{b})}, \qquad \frac{C_{\mathrm{A}}}{2^{3/2}\sqrt{I_b}} = C_b.$$

Indeed, $\overline{F}_{\chi_4}(\sqrt{2uI_b})(2uI_b)^{-3/2} C_{\mathrm{A}} \underset{u\to\infty}{\sim} C_b u^{-1/2} e^{-uI_b}$. Thus the angular calculation recovers (C.10). A negative H_1 or H_2 requires $t > 2s$ and the two cannot both be negative, whereas $\overline{H} > 0$ throughout the interval.

Put $A_u = \{Y(s) > u, Y(t) > u\}$. On $\{Y(s) > u\}$ write $\Theta_u = B(s)/\|B(s)\|$, $\Theta_u^{\perp} = (-\Theta_{u,2}, \Theta_{u,1})$ and $N = B(t) - B(s)$. Put

$$Z_{\parallel,u} = \frac{\langle \Theta_u, N\rangle}{\sqrt{\Delta}}, \qquad Z_{\perp,u} = \frac{\langle \Theta_u^{\perp}, N\rangle}{\sqrt{\Delta}}.$$

Corollary C.3. *Let E, Z_+, Z_0 be independent with E exponential of mean $2s$, Z_+ standard half-normal and Z_0 standard normal. Then*

$$\big(Y(s)-u, Z_{\|,u}, Z_{\perp,u}\big) \;\Big|\; A_u \xrightarrow[u\to\infty]{d} (E, Z_+, Z_0), \tag{C.12}$$

and

$$\left(Y(s)-u, \frac{Y(t)-u}{2\sqrt{u\Delta}}\right) \;\Big|\; A_u \xrightarrow[u\to\infty]{d} (E, Z_+), \tag{C.13}$$

$$\min(Y(s), Y(t)) - u \;\Big|\; A_u \xrightarrow[u\to\infty]{d} E. \tag{C.14}$$

Remark C.4. *The constant C_b diverges at both ends of $1<b<(t/s)^2$. The transition at either endpoint has scale $u^{-1/2}$ in b. At the lower endpoint, for $b_u = 1+\beta u^{-1/2}$ and fixed $\beta\in\mathbb{R}$, the increment representation yields*

$$\frac{\mathbb{P}\{Y(s)>u, Y(t)>b_u u\}}{\mathbb{P}\{Y(s)>u\}} \xrightarrow{u\to\infty} \overline{\Phi}\left(\frac{\beta}{2\sqrt{\Delta}}\right)$$

where $\overline{\Phi}$ is the standard normal survival function.

Remark C.5. *Let $\Delta_u = t_u - s > 0$ tend to zero as $u\to\infty$ and put*

$$p_{1,t_u}(u) = \mathbb{P}\{Y(s)>u, Y(t_u)>u\}.$$

If $u\Delta_u \to c\in[0,\infty)$ then, conditional on $Y(s)>u$, the second constraint converges to $E+2\sqrt{c}Z>0$ with independent $E\sim \mathrm{Exp}(1/(2s))$ and $Z\sim N(0,1)$. Since

$$\mathbb{P}\big\{E+2\sqrt{c}Z>0\big\} = \frac{1}{2} + \int_{-\infty}^{0} e^{\sqrt{c}z/s}\frac{e^{-z^2/2}}{\sqrt{2\pi}}\mathrm{d}z = \frac{1}{2} + \exp\left(\frac{c}{2s^2}\right)\overline{\Phi}\left(\frac{\sqrt{c}}{s}\right),$$

we obtain

$$\frac{p_{1,t_u}(u)}{\mathbb{P}\{Y(s)>u\}} \xrightarrow{u\to\infty} \frac{1}{2} + \exp\left(\frac{c}{2s^2}\right)\overline{\Phi}\left(\frac{\sqrt{c}}{s}\right). \tag{C.15}$$

For $u\Delta_u\to 0$ the limit in (C.15) is one. For $u\Delta_u\to\infty$ division of the conditional constraint by $2\sqrt{u\Delta_u}$ yields $Z>0$ in the limit and hence the factor $1/2$ in (C.11).

C.3. **Proofs.**

Proof of Theorem C.1. The Gaussian energy

$$\mathcal{I}(x,y) = \frac{\|x\|^2}{2s} + \frac{\|y-x\|^2}{2\Delta} \tag{C.16}$$

is least at fixed norms when x, y are parallel with the same direction. Scaling both norms by $\sqrt{u}$ therefore yields

$$\min_{r\ge 1,\ \rho\ge\sqrt{b}} \left(\frac{r^2}{2s} + \frac{(\rho-r)^2}{2\Delta}\right). \tag{C.17}$$

At $(1,\sqrt{b})$ the two partial derivatives are

$$\mu_r = \frac{t-s\sqrt{b}}{s\Delta}, \qquad \mu_\rho = \frac{\sqrt{b}-1}{\Delta}. \tag{C.18}$$

By strict convexity the minimiser is $(1,1)$ for $b\le 1$, $(1,\sqrt{b})$ for $1<b<(t/s)^2$ and $((s/t)\sqrt{b},\sqrt{b})$ for $b\ge(t/s)^2$. The minimum in (C.17) is thus

$$\begin{cases} 1/(2s), & b\le 1,\\ I_b, & 1<b<(t/s)^2,\\ b/(2t), & b\ge(t/s)^2. \end{cases} \tag{C.19}$$

For $1 < b < (t/s)^2$, integration of the Gaussian increment over its polar angle yields the joint density of $(Y(s), Y(t))$

$$f_{s,t}(x,z) = \frac{1}{4s\Delta} \exp\left(-\frac{x}{2s} - \frac{x+z}{2\Delta}\right) I_0\left(\frac{\sqrt{xz}}{\Delta}\right), \qquad x, z > 0. \tag{C.20}$$

The Bessel expansion [14, Eq. 10.40.1]

$$I_0(v) = \frac{e^v}{\sqrt{2\pi v}}(1 + O(v^{-1})), \qquad v \to \infty, \tag{C.21}$$

applies uniformly on $\xi \geq 1, \eta \geq b$ since $u\sqrt{\xi\eta}/\Delta \geq u\sqrt{b}/\Delta$. Hence

$$u^2 f_{s,t}(u\xi, u\eta) = \frac{u^{3/2}(\xi\eta)^{-1/4}}{4s\sqrt{2\pi\Delta}} e^{-u\mathcal{J}(\xi,\eta)}(1 + O(u^{-1})) \tag{C.22}$$

with

$$\mathcal{J}(\xi,\eta) = \frac{\xi}{2s} + \frac{(\sqrt{\eta} - \sqrt{\xi})^2}{2\Delta}. \tag{C.23}$$

The corner slopes are

$$A := \mathcal{J}_\xi(1,b) = \frac{t - s\sqrt{b}}{2s\Delta} > 0, \qquad D := \mathcal{J}_\eta(1,b) = \frac{\sqrt{b} - 1}{2\Delta\sqrt{b}} > 0. \tag{C.24}$$

Concavity of $\sqrt{\xi\eta}$ makes $\mathcal{J}$ convex. At $(\xi,\eta) = (1 + x/u, b + y/u)$ we thus have $u(\mathcal{J}(\xi,\eta) - I_b) \geq Ax + Dy$ with equality in the limit for fixed $x, y \geq 0$. Since $(\xi\eta)^{-1/4} \leq b^{-1/4}$, dominated convergence in (C.22) yields

$$p_b(u) \underset{u\to\infty}{\sim} \frac{b^{-1/4} u^{-1/2} e^{-uI_b}}{4s\sqrt{2\pi\Delta}} \int_0^\infty \int_0^\infty e^{-Ax - Dy} \mathrm{d}x\mathrm{d}y = C_b u^{-1/2} e^{-uI_b},$$

where (C.24) yields (C.10).

For $b = 1$ write $U = (\sqrt{1 - a^2 - \gamma^2} v, av + \gamma v^\perp)$ near (C.7). With $q_i = G_i(U)^{-1/2}$ and $q_* = s^{-1/2}$, (C.5)–(C.6) yield

$$q_1/q_* = (1 - a^2 - \gamma^2)^{-1/2} = 1 + \tfrac{1}{2}(a^2 + \gamma^2) + o(a^2 + \gamma^2), \tag{C.25}$$

$$q_2/q_* = 1 - \sqrt{\Delta/s}\, a + O(a^2 + \gamma^2). \tag{C.26}$$

Thus Theorem 6.1(W) applies with $c = 2$ and $H_s = I_2$. The circle has length 2π and the angular density is $1/(2\pi^2)$, whence $C_{\mathrm{W}} = 1$. Since $r_\tau^2 = \lambda_\tau = u/s$ and $\overline{F}_{\chi_4}(r) = e^{-r^2/2}(1 + r^2/2)$, the second line of (C.8) follows.

For $b < 1$, conditionally on $Y(s) > u$, memorylessness yields $Y(s) = u + E_s$ with E_s exponential of mean $2s$. The independent increment in (C.4) implies $Y(t)/u \to 1$ in conditional probability. For $b \geq (t/s)^2$ use the regression $B(s) = (s/t)B(t) + W$ with $W \sim N_2(0, (s\Delta/t)I_2)$ independent of $B(t)$. Given $Y(t) > bu$ we have $Y(t) = bu + E_t$ with E_t exponential of mean $2t$. Hence $Y(s)/u \to (s/t)^2 b$ in conditional probability. At equality $b = (t/s)^2$, under the same conditioning,

$$\frac{Y(s) - u}{2\sqrt{us\Delta/t}} = \frac{(s/t)^2 E_t + 2(s/t)\langle B(t), W\rangle + \|W\|^2}{2\sqrt{us\Delta/t}} \xrightarrow[u\to\infty]{d} N(0,1)$$

by isotropy of W and $Y(t)/u \to b$. The strict cases therefore have factor one and the upper boundary has factor $1/2$. Since $\mathbb{P}\{Y(r) > v\} = e^{-v/(2r)}$ for $r, v > 0$, this proves (C.8). □

Proof of Corollary C.3. Given $Y(s) > u$, memorylessness and isotropy of the independent increment allow us to realise $(Y(s) - u, Z_{\parallel,u}, Z_{\perp,u})$ as independent (E, Z_1, Z_2) with E as stated and Z_1, Z_2 standard normal. On this realisation

$$D_u := \frac{Y(t) - u}{2\sqrt{u\Delta}} = \sqrt{1 + E/u}\, Z_1 + \frac{E + \Delta(Z_1^2 + Z_2^2)}{2\sqrt{u\Delta}} \xrightarrow[u\to\infty]{\text{a.s.}} Z_1.$$

Thus $\mathbf{1}_{\{D_u>0\}} \to \mathbf{1}_{\{Z_1>0\}}$ almost surely. Dominated convergence in the conditional expectation of any bounded continuous test function, with denominator $\mathbb{P}\{D_u > 0\} \to 1/2$, yields (C.12) and (C.13). Since $Z_+ > 0$ almost surely, $Y(t) - u$ diverges in conditional probability while $Y(s) - u$ is tight, proving (C.14). □

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